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BACHELOR THESIS

**Numerical model of a turbojet propulsion control system
used in a cruise missile**

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Abstract

The progressive development of aerospace propulsion systems, encompassing both high-power or high-thrust units and those dedicated to small, often unmanned aerial vehicles (UAVs), necessitates the development and implementation of new technologies, or the verification and optimization of existing ones, related to the broad exploitation of these technical systems. One of the most important segments currently associated with their use, stemming from their increasing automation and digitization, are automatic control and regulation systems and their accompanying computer models. The former ensure deliberate, safe, and as fully controllable as possible physical operation of the aforementioned equipment within defined operational ranges, while the latter provide instrument programmability, serving also as a valuable tool for conducting simulations, tests, and training operating personnel considering the specifics of the given object and the operational requirements of so-called digital twins. Among the intensely developing aerospace systems currently are cruise missiles, relatively small observation UAVs (including civilian ones), and aerial target drones. These objects, gaining popularity due to the global geopolitical situation and changing economic needs, represent interesting cases, often diverging in terms of complexity, technical standards, and operational range—from more commonly used manned aircraft (also regarding control or regulation techniques), which have larger take-off masses or a limited role of propulsion in shaping the aerodynamic phenomena of the carrier. Among them are systems utilizing turbojet engines. The aim of this work is to create a numerical, zero or one dimensional, appropriately linearized, engine used in modern cruise missiles, and to associate it with a self developed computer model, serving for multi-variant analyses of relationships between the most important input and output signals, also in terms of common disturbances, within the aforementioned engine (with an appropriate degree of automation and considering the nature of the engine itself).

Increasing the degree or range of controllability of this class of aircraft engines, achieved, among other things, based on the titular project, may lead to a future reduction in the degree of hybridization of propulsion systems used in this class of aerial objects, an increase in their operational range, or the unification of engineering and technical processes related to the selection of ACS for the titular UAVs. The model itself could also be used, for example, during the training of engineering personnel or for analysing issues related to rocket and missile aerodynamics.

Finally, the effectiveness of the proposed solution will be verified through the simulation of the complete closed-loop control system. This stage involves the design and implementation of two distinct control strategies: a classical Proportional-Integral (PI) controller, serving as a reference baseline, and the developed Model Predictive Control (MPC) algorithm. A comparative analysis of both approaches will be conducted to evaluate their performance regarding trajectory tracking, reaction time, and robustness against disturbances.

Streszczenie

Postępujący rozwój lotniczych układów napędowych, obejmujący zarówno jednostki o dużej mocy i ciągu, jak i silniki przeznaczone dla małych, często bezzałogowych statków powietrznych (UAV), wymusza opracowywanie nowych technologii oraz weryfikację i optymalizację rozwiązań istniejących. Jednym z kluczowych obszarów ich współczesnej eksploatacji, wynikającym z rosnącej automatyzacji i cyfryzacji, są automatyczne układy sterowania i regulacji (ACRS) oraz towarzyszące im modele komputerowe. Układy te zapewniają bezpieczną i kontrolowalną pracę silników w określonych zakresach operacyjnych, natomiast modele komputerowe umożliwiają prowadzenie symulacji, badań oraz szkolenie personelu w kontekście koncepcji cyfrowych bliźniaków.

Do intensywnie rozwijających się obecnie systemów lotniczych należą m.in. pociski manewrujące, niewielkie bezzałogowe statki powietrzne o charakterze obserwacyjnym oraz powietrzne cele treningowe. Obiekty te, zyskujące na znaczeniu w związku z globalną sytuacją geopolityczną i zmieniającymi się potrzebami ekonomicznymi, często różnią się pod względem złożoności, standardów technicznych i metod sterowania od statków powietrznych załogowych. Wśród nich znajdują się również systemy wykorzystujące silniki turboodrzutowe.

Celem niniejszej pracy jest opracowanie numerycznego, zero- lub jednowymiarowego, odpowiednio zlinearyzowanego modelu silnika stosowanego we współczesnych pociskach manewrujących oraz powiązanie go z autorskim modelem komputerowym. Model ten przeznaczony jest do wielowariantowej analizy zależności pomiędzy kluczowymi sygnałami wejściowymi i wyjściowymi, z uwzględnieniem typowych zakłóceń oraz stopnia automatyzacji i specyfiki analizowanej jednostki napędowej.

Zwiększenie zakresu sterowalności tej klasy silników może w przyszłości przyczynić się do redukcji stopnia hybrydyzacji układów napędowych, zwiększenia zasięgu operacyjnego UAV oraz ujednolicenia procesów inżynierskich związanych z doбором układów ACRS. Opracowany model może być również wykorzystany w szkoleniu personelu technicznego oraz w analizach zagadnień związanych z aerodynamiką rakiet i pocisków.

Skuteczność zaproponowanego rozwiązania zostanie zweryfikowana poprzez symulację kompletnego układu sterowania w pętli zamkniętej. W tym celu zaprojektowane i zaimplementowane zostaną dwie strategie sterowania: klasyczny regulator PI oraz algorytm sterowania predykcyjnego MPC. Oba podejścia zostaną porównane pod względem dokładności śledzenia trajektorii, czasu reakcji oraz odporności na zakłócenia.

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Table of Symbols

H	Altitude
T_H	Temperature at altitude H
p_H	Pressure at altitude H
T_1^*	Total temperature at compressor inlet
p_1^*	Total pressure at compressor inlet
$\dot{m}$	Air flow
Ma	Mach number
σ_{in}	Inlet efficiency
T_2^*	Total temperature at compressor exit
p_2^*	Total pressure at compressor exit
c_p	heat capacity of air
κ	Adiabatic index
$\dot{m}_f$	Fuel flow
W	Lower heating value of jet fuel
ζ_{CC}	Combustion efficiency coefficient
σ_{CC}	Combustion chamber efficiency
T_3^*	Total temperature at combustion chamber exit
p_3^*	Total pressure at combustion chamber exit
κ'	Adiabatic index of exhaust gas
c_p'	Heat capacity of exhaust gas
K	Thrust
c_8	Exhaust gas speed at nozzle exit
c_H	Ambient air flow speed
π_n	Nozzle pressure ratio
π_{cr}	Critical nozzle pressure ratio
p_8	Static pressure at nozzle exit
J	Shaft moment of inertia
T_T	Turbine torque
T_C	Compressor torque
n_{nom}	Nominal rotational speed of engine shaft

1 Introduction

1.1 Propulsion system in contexts of cruise missile

In modern civil aviation, the dominant propulsion solution is the high-bypass ratio turbofan engine. These state-of-the-art constructions provide high thrust at low fuel consumption, making them robust, reliable, and optimal for long-distance subsonic flights. However, the operational requirements for cruise missiles and high-speed unmanned aerial vehicles (UAVs) differ significantly from commercial aviation, making high-bypass engines unsuitable for these applications.

For cruise missiles, the priority shifts towards high, even supersonic speed, compact dimensions, and mechanical simplicity. In this specific use case, the turbojet engine remains the superior choice. The single spool turbojet offers a much smaller frontal cross-section compared to turbofans, which drastically reduces aerodynamic drag, a critical factor at high Mach numbers. Furthermore, the absence of a large bypass fan allows for a simpler, lighter, and more compact construction. The spectrum of propulsion systems used is shown in the Table 1

From an economic perspective, cruise missiles are often expendable assets designed for "one way" missions. Therefore, the propulsion system must balance performance with manufacturing costs. The relative simplicity of a single-spool jet engine makes it a cost-effective solution for missile propulsion systems, providing the necessary high thrust-to-weight ratio without the complexity and expense of multi-spool commercial engines.

Table 1: Cruise missiles propulsion system overview [12]

Missile	Propulsion type	Thrust	Max speed
AGM-28 Hound Dog	Turbojet (Pratt & Whitney J52-P-3)	33 kN	Mach 2.1
Apache	Turbojet (Microturbo TRI 60-30)	5.4 kN	1000 km/h
Storm Shadow / SCALP	Turbojet (Microturbo TRI 60-30)	5.4 kN	Mach 0.95
AGM-158 JASSM	Turbojet (Teledyne CAE J402)	3.0 kN	Mach 0.85
SOM	Turbojet (Microturbo TRI 40)	2.5 kN	Mach 0.94
Atmaca	Turbojet (Kale KTJ-3200)	3.2 kN	Mach 0.90
MdCN	Turbojet (Microturbo TR50)	3.4 kN	800 km/h
3M-54 Kalibr	Multi-stage Rocket/Turbojet	2.7 kN	Mach 2.9
Tomahawk (BGM-109)	Turbofan (Williams F107)	2.7 kN	880 km/h
AGM-86 ALCM	Turbofan (Williams F107)	2.7 kN	Mach 0.73

1.2 Market overview

The market for small-scale turbojet engines often referred to as micro-turbines is currently experiencing rapid growth, driven primarily by the expanding defense and aerospace sectors.

While the commercial aviation market focuses on fuel efficiency and noise reduction, the tactical missile and UAV market prioritizes thrust density, reliability, and rapid production capabilities.

Recent geopolitical conflicts have highlighted the strategic importance of loitering munitions, target drones, and cruise missiles. This has created a surge in demand for propulsion units in the thrust class of 300 N to several kilo-newtons. Manufacturers are increasingly moving towards full authority digital engine control systems (FADEC) even in smaller engines to improve performance and fuel economy during complex flight maneuvers [3].

Moreover, the trend towards swarming tactics implies the need for mass-produced, low-cost propulsion systems. Consequently, the industry is witnessing a shift towards advanced manufacturing techniques, such as additive manufacturing such as 3D printing, to reduce the number of parts and lower the production cost of engine components like compressor wheels and combustion chambers. This thesis fits into this modern context by exploring control strategies that can further optimize the performance of such compact propulsion systems.

To illustrate the practical application of turbojet engines in modern weapon systems, it is beneficial to examine specific propulsion units that dominate the market. Despite the high efficiency of turbofans, the turbojet cycle remains the optimal solution for many tactical applications due to its compact size and high specific thrust. Two dominant engine families in the 3 – 6 kN thrust class serve as prime examples of this technology's relevance.

A prominent example of industry standards is the Teledyne CAE J402 engine Fig. 1. Originally developed in the 1970s, it has been continuously modernized and is characterized by its sealed round design, allowing for long-term storage without maintenance. This engine powers the AGM-84 Harpoon, an over-the-horizon anti-ship missile. Significantly, the J402-CA-100 variant was also selected for the initial versions of the modern AGM-158 JASSM (Joint Air-to-Surface Standoff Missile). The choice of a turbojet for this stealth platform highlights its critical advantage: a smaller cross-sectional area compared to turbofans, which facilitates internal carriage and reduces the missile's radar signature.

In the European market, the Safran Microturbo TRI 60 family represents a widely adopted solution, renowned for its simple single-spool architecture. This engine is designed to operate efficiently across a wide flight envelope, including high-subsonic speeds. It serves as the propulsion unit for the Storm Shadow / SCALP EG, a low-observable cruise missile capable of delivering a 450 kg payload over ranges exceeding 500 km. Another notable application is the RBS-15 Mk3 heavy anti-ship missile, which utilizes the TRI 60 to achieve high maneuverability and Mach 0.9 speeds during the terminal engagement phase.

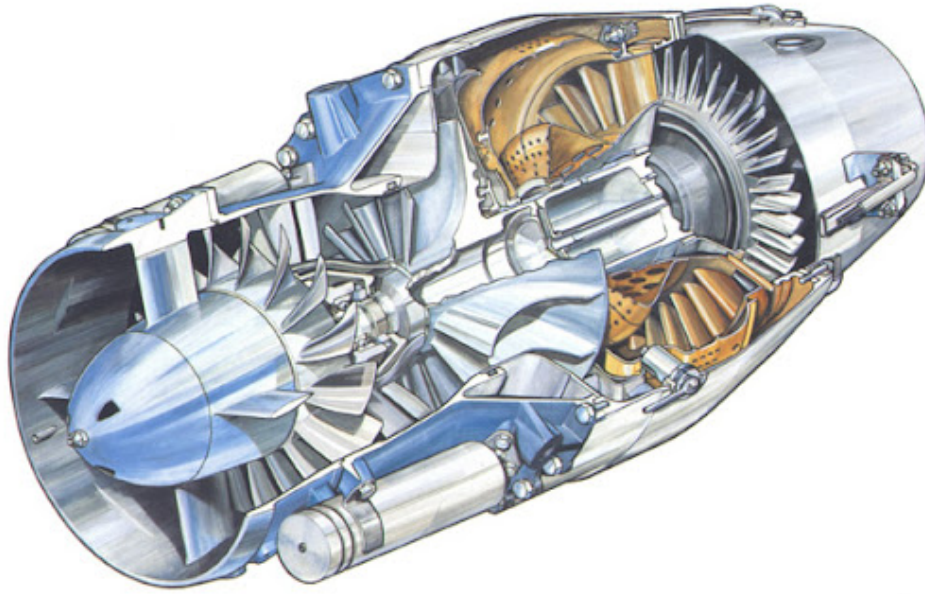


Figure 1: Cutaway view of the Teledyne CAE J402-CA-400 turbojet engine. [1]

These examples validate the industrial relevance of the modeled object. They demonstrate that for tactical ranges where the engine operates for a relatively short duration, the turbojet's compactness and cost-effectiveness outweigh the fuel efficiency benefits of bypass engines.

1.3 Numerical models

The development of modern propulsion systems is inseparably linked to numerical modeling. In the aerospace industry, computer simulations allow engineers to predict engine behavior, optimize performance, and design control algorithms without the prohibitive cost and risk associated with physical prototyping. Numerical models of gas turbine engines can be categorized based on their high fidelity and computational complexity, ranging from three-dimensional Computational Fluid Dynamics (CFD) to zero-dimensional (0-D) lumped parameter models.

At the highest level of fidelity lie 3-D CFD simulations. These models solve the Navier-Stokes equations to provide a detailed analysis of the airflow through complex geometries, such as compressor blades or turbine stators. While indispensable for the aerodynamic design of individual components, CFD models are computationally expensive, often requiring hours or days to simulate a fraction of a second of engine operation. Consequently, their high computational burden makes them unsuitable for system-level dynamic analysis or real-time control applications, where calculation speed is paramount.

For the purpose of performance synthesis and control system design, 0-D (zero-dimensional) or Component Level Models (CLM) are the industry standard. In this approach, the engine is treated not as a continuous fluid volume, but as a network of functional modules: intake, compressor, combustor, turbine, and nozzle.

The development of propulsion systems relies heavily on numerical modeling to reduce costs and risks associated with physical testing. Models can be categorized based on their dimensionality and time-dependency. As described by [11], models are generally divided into static (steady-state) components, which are time-independent, and dynamic models, which account for transient states and time-variable parameters.

For the purpose of control system design, zero-dimensional (0-D) lumped parameter models are the industry standard. This approach, utilized in NASA's T-MATS (Toolbox for the Modeling and Analysis of Thermodynamic Systems) [5], represents the engine as a series of functional blocks (compressor, burner, turbine) where physics-based thermodynamic equations are solved iteratively. The primary advantage of this method is the balance between high fidelity and computational efficiency, allowing for real-time simulation.

Alternative approaches exist, such as data-driven models utilizing Neural Networks [14]. Demonstrated that engine subsystems can be modeled using algebraic relationships derived from experimental rig tests and mapped via neural networks to predict performance. While effective for specific operating points, physics-based 0-D models like the one developed in this thesis offer greater flexibility for analyzing fundamental thermodynamic changes and are preferred for initial control law development.

1.4 Control theory

The control of gas turbine engines has evolved from simple hydro mechanical governors to sophisticated Full Authority Digital Engine Control (FADEC) systems. To implement modern control strategies, such as Model Predictive Control (MPC), a mathematical representation of the system in the state-space form is required. Since the thermodynamic behavior of a turbojet engine is highly non-linear, linear models must be derived for specific operating points. As noted by [13], the difficulty in developing control systems lies in converting non-linear cycle decks into accurate linear state-space models that capture rotor dynamics and heat soak effects. For small-scale turbojet engines, which are the focus of this study, the state-space representation is particularly effective. A linear mathematical model for a small turbojet [4] with a variable exhaust nozzle, identifying rotor speed (n) and internal pressures (p_3 , p_6) as critical state variables representing the energy storage of the system. This thesis utilizes the Beta-lines method common in T-MATS simulations. Therefore, the state vector will incorporate the Beta parameter to explicitly monitor the surge margin, as detailed in Chapter 3. Linearized state-space model to predict future system states, which is the foundational requirement for the MPC algorithm described in Chapter 4.

2 Reference model

The following chapter details the mathematical formulation of the single-spool turbojet engine reference model. The modeling strategy adopted in this thesis is based on the thermodynamic analysis of the Brayton-Joule cycle Fig. 2, which describes the fundamental operation of a gas turbine engine consisting of compression, combustion, and expansion phases. For the purpose of model computing air flow is considered as ideal gas.

To represent the actual physical behavior of the propulsion system, the model transitions from an ideal cycle to a real cycle representation by incorporating component isentropic efficiencies and pressure losses. The engine is treated as a modular system of zero-dimensional (0-D) functional blocks (Component Level Model - CLM), where the flow properties are averaged across the cross-section of each engine station shown in Fig.3.

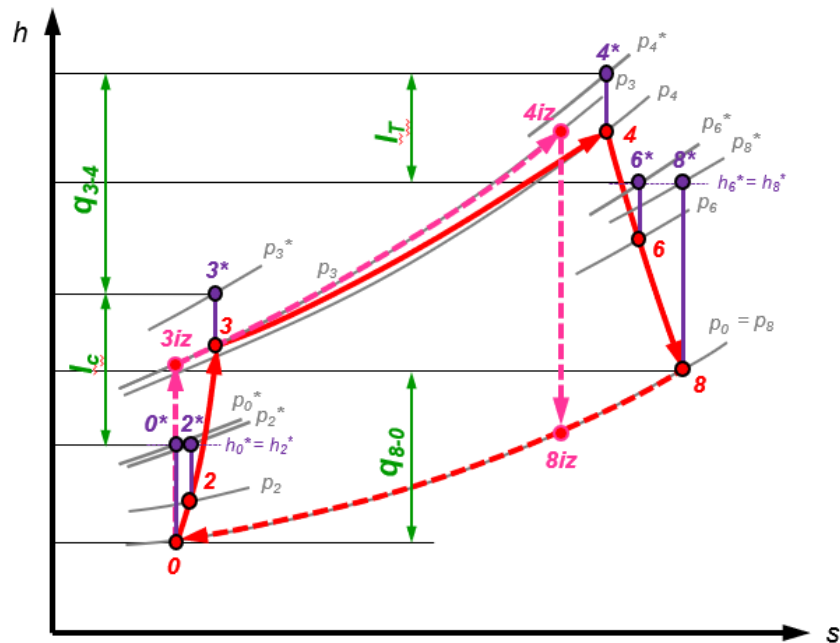


Figure 2: Brayton-Joule thermodynamic cycle for a turbojet in T-s reference frame.

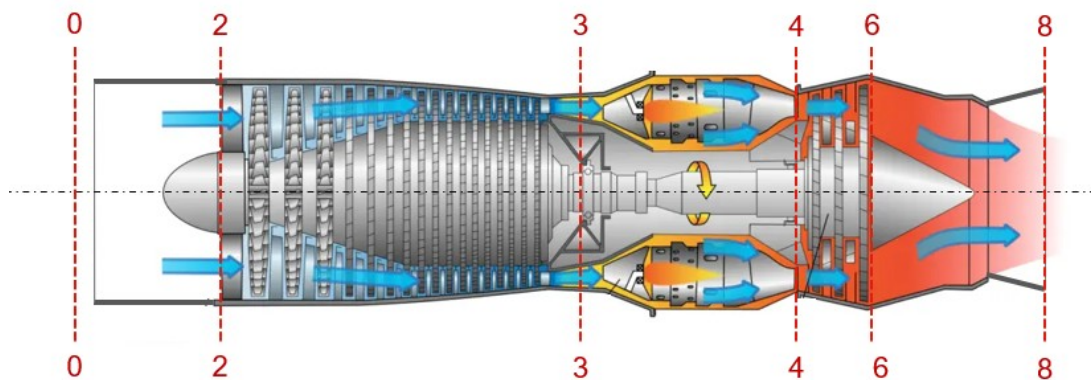


Figure 3: Turbojet cross-section.

The mathematical description of the system relies on three fundamental physical principles:

- **Conservation of Mass:** Ensuring flow continuity between components.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (2.1)$$

- **Conservation of Energy:** Balancing the enthalpy changes across the compressor, combustor, and turbine according to first law of thermodynamics.

$$\delta Q = dU + \delta W, \quad (2.2)$$

- **Conservation of Momentum:** Determining the thrust generation in the nozzle and the torque balance on the shaft implied by Newton's laws of motion.

$$\frac{d}{dt}(p_1 + p_2) = 0. \quad (2.3)$$

While the thermodynamic states at each station are calculated using algebraic relationships and empirical performance maps, the transient behavior of the engine is captured through differential equations describing the mechanical dynamics of the rotor shaft.

2.1 Environment

Ambient conditions vary with the altitude at which the engine is operating. For computational purposes, International Standard Atmosphere (ISA) has been used to calculate the temperature, air pressure and sound speed at operating altitude. At the sea-level according to ISA (Fig. 4) $T_0 = 288,15K$, $p_0 = 1013250Pa$, $a_0 = 340,3\frac{m}{s}$. To calculate conditions at given altitude in the range from 0 to 11km, the following equations have been used:

$$T_H = T_0 - 6,5 \frac{K}{km} \quad (2.4)$$

$$p_H = p_0 \left(1 - \frac{H}{44300}\right)^{5,2567} \quad (2.5)$$

$$a_H = a_0 \sqrt{\frac{T_H}{T_0}} \quad (2.6)$$

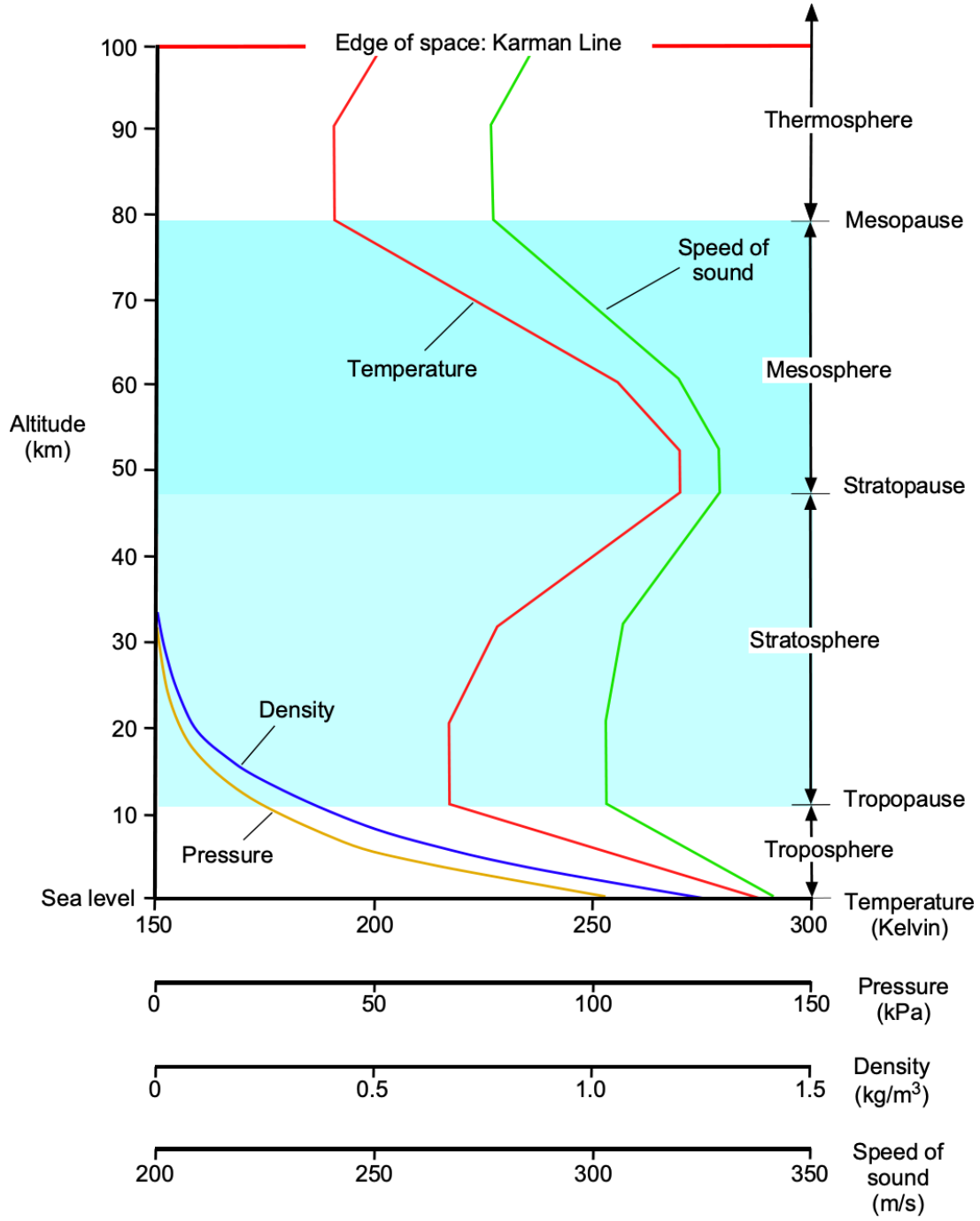


Figure 4: International standard atmosphere [9].

2.2 Inlet

Engine inlet defines air flow conditions at compressor first stage. For model purposes inlet is treated as isolated channel (adiabatic), temperature and pressure of air flow exiting inlet has been calculated:

$$T_1^* = T_H^* = T_H \left(1 + \frac{\kappa - 1}{2} Ma_H^2 \right) \quad (2.7)$$

$$p_1^* = p_H^* \sigma_{wl} = \sigma_{wl} p_H \left(1 + \frac{\kappa - 1}{2} Ma_H^2\right) \quad (2.8)$$

Loss of air pressure σ_{in} can be approximated by polynomial: [11]

$$\begin{aligned} \sigma_{in} = & 0,97 - 0,02241(Ma_{in} - 1)^2 - 0,14561(Ma_{in} - 1)^2 + \\ & + 0,086282(Ma_{in} - 1)^4 - 0,014342(Ma_{in} - 1)^5 \end{aligned} \quad (2.9)$$

2.3 Compressor

The compressor is modeled as a component that increases the total pressure of the air flow driven by the mechanical power provided by turbine and transferred by shaft. Due to the complex aerodynamic phenomena occurring within the blade rows, the compressor behavior is non-linear and is typically represented by performance maps. In this thesis, to avoid complex numerical methods associated with compressor characteristic, the performance maps are implemented using the Beta-lines (β) method, as recommended in turbojet engines simulation [5], [11]. In this approach, the compressor characteristics are parameterized using an auxiliary coordinate β (where $\beta \in [0, 1]$), which intersects the constant speed lines. Consequently, the corrected compressor pressure ratio ($\pi_{c,corr}$), corrected mass flow ($\dot{m}_{c,corr}$), and corrected isentropic efficiency ($\eta_{c,corr}$) become functions of the corrected rotational speed (n_{corr}) and the β coordinate:

$$\pi_{c,corr} = f(n_{corr}, \beta) \quad (2.10)$$

$$\dot{m}_{c,corr} = f(n_{corr}, \beta) \quad (2.11)$$

$$\eta_{c,corr} = f(n_{corr}, \beta) \quad (2.12)$$

The corrected rotational speed is defined to account for varying inlet conditions:

$$n_{corr} = \frac{n}{\sqrt{T_1^*/T_{ref}}} \quad (2.13)$$

Vital part of β - lines method is beta itself, to implement computational solver like shown in [11]. equation has been derived:

$$\beta = \int \frac{\dot{m}_c - \dot{m}_{ref}}{\dot{m}_{nom}} \quad (2.14)$$

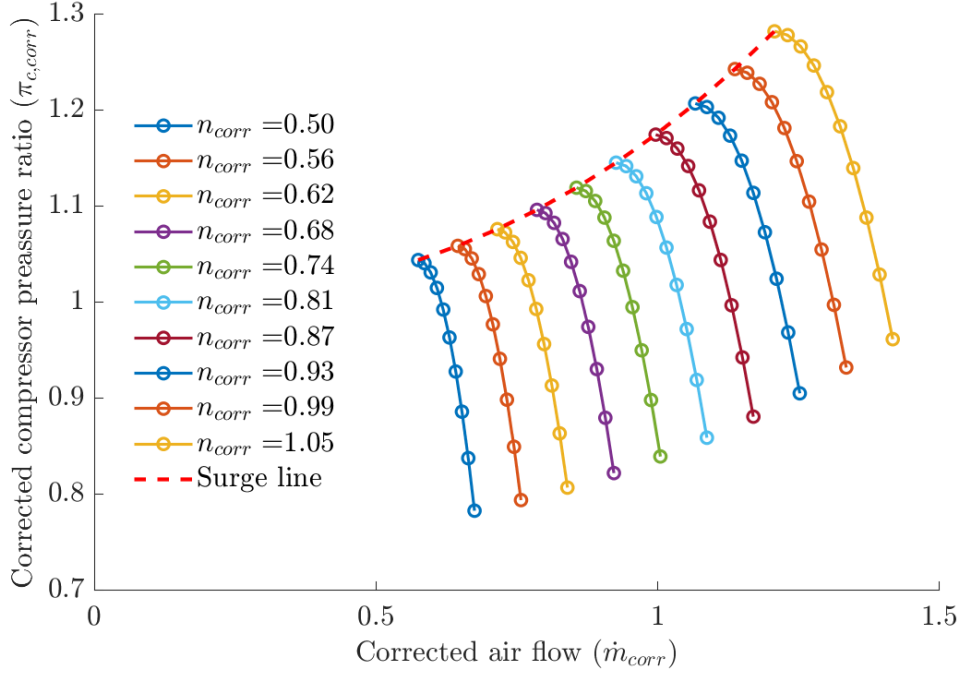


Figure 5: Compressor corrected pressure ratio characteristic.

With passing simulation steps beta due to integration will converge to steady-state compressor operational point.

The generated performance maps used in the simulation are presented in Fig. 5 and Fig. 6.

The pressure ratio characteristics is pictured on Fig.5, clearly showing the surge line limit (left boundary) and the operational range. Fig.6 depicts the isentropic efficiency.

To obtain physical values for the simulation, the dimensionless map outputs are scaled by the design point parameters (nominal mass flow $\dot{m}_{nom}$, nominal pressure ratio π_{nom} , nominal compressor efficiency $\eta_{c,nom}$):

$$\begin{aligned}\dot{m}_{c,corr} &= \dot{m}_{map} \cdot \dot{m}_{nom} \\ \pi_c &= \pi_{c,map} \cdot \pi_{nom} \\ \eta_c &= \eta_{c,map} \cdot \eta_{c,nom}\end{aligned}\tag{2.15}$$

Further more true air flow has to be calculated:

$$\dot{m}_c = \dot{m}_{c,corr} \frac{p_1/p_{ref}}{\sqrt{T_1/T_{ref}}}\tag{2.16}$$

Values for above-mentioned variables are placed in Table 2.

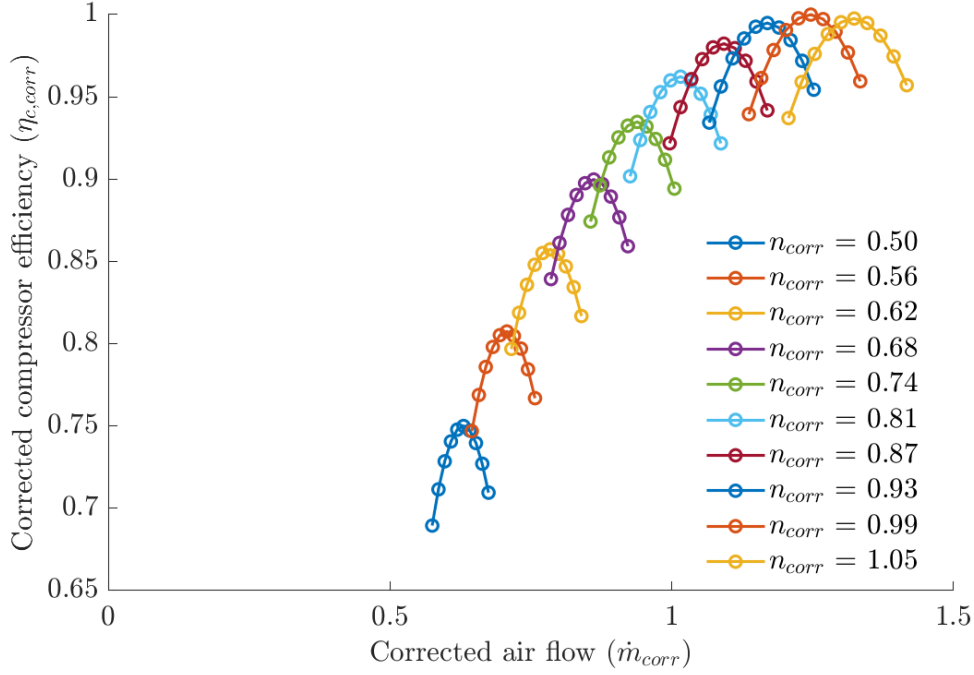


Figure 6: Compressor corrected efficiency characteristic.

Table 2: Compressor characteristic variables

Variable name	Value	Unit
nominal mass flow $\dot{m}_{nom}$	4.5	kg/s
nominal pressure ratio π_{nom}	5.8	-
nominal compressor efficiency $\eta_{c,nom}$	0.78	-
reference pressure p_{ref}	101325	Pa
reference temperature T_{ref}	288.15	K

In Matlab/SIMULINK environment compressor characteristics are represented as 2D-Lookup Table. In order to visualize model implementation, maps are presented as 3D surfaces on figures (Fig.7, Fig.8, Fig.9).

$$T_2^* = T_1^* \left(1 + \frac{\pi_s^{*\frac{\kappa-1}{\kappa}} - 1}{\eta_s^*} \right) \quad (2.17)$$

Pressure gain is derived from compressor pressure ratio (π_c):

$$p_2^* = p_1^* \pi_c^* \quad (2.18)$$

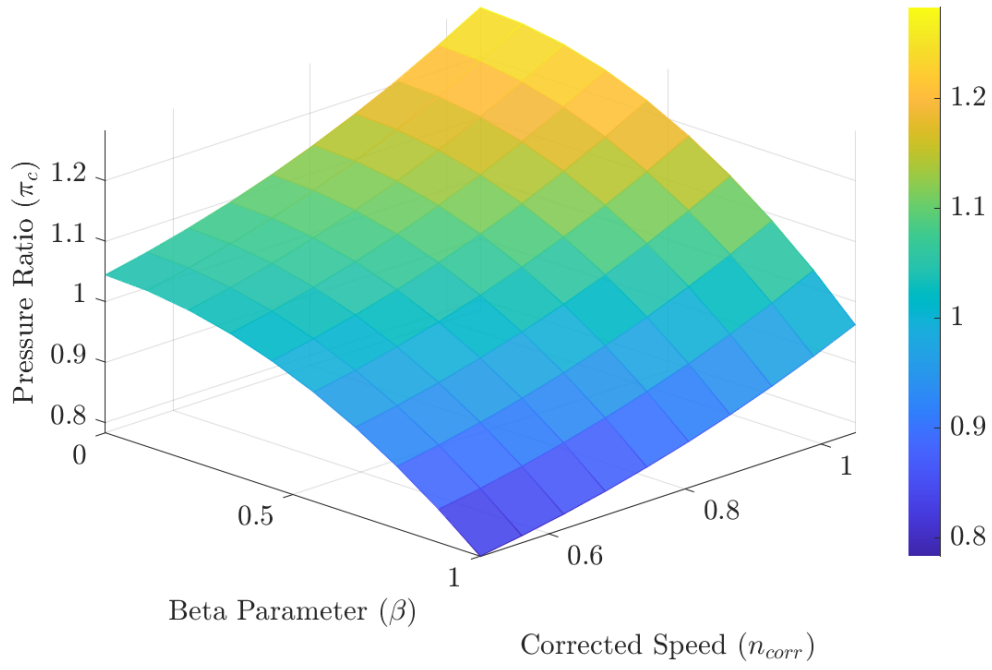


Figure 7: 3D surface of compressor corrected pressure ratio characteristic.

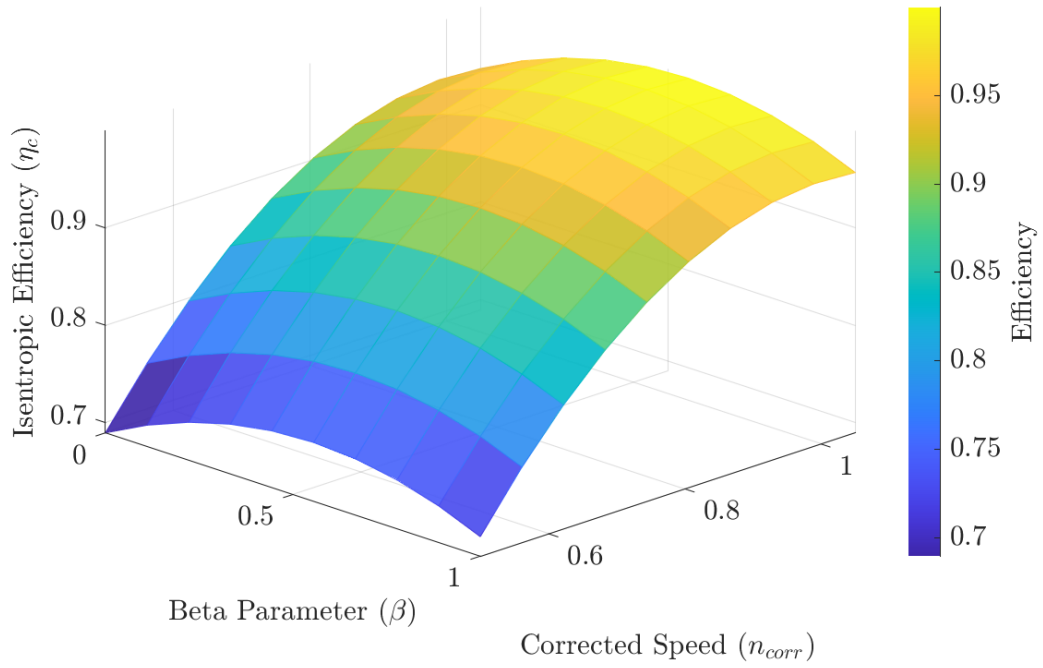


Figure 8: 3D surface of compressor corrected efficiency characteristic.

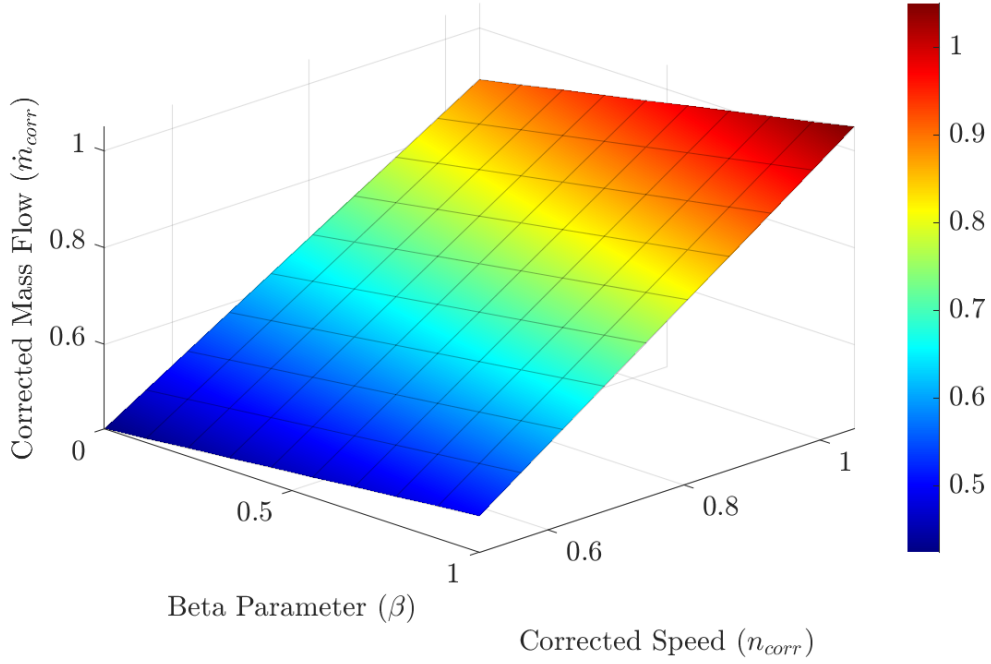


Figure 9: 3D surface of compressor corrected air flow characteristic.

The temperature rise across the compressor is calculated based on the definition of isentropic efficiency:

By combining equations (2.7, 2.17) power of compressor can be calculated:

$$P_C = c_p \dot{m} (T_2^* - T_1^*) \quad (2.19)$$

2.4 Combustion chamber

In combustion chamber fuel is burned for the purpose of increasing heat of pressurized air. As a result of additional heat, enthalpy is also rising to be used in turbine to produce work. Considering combustion chamber is not a heat volume enthalpy on CC exit can be calculated from following equation:

$$i_3^* = \frac{W_u \zeta_{CC} \dot{m}_f + i_2^* \dot{m}}{\dot{m} + \dot{m}_f} = \frac{W_u \zeta_{CC} \tau + i_2^*}{1 + \tau} \quad (2.20)$$

where:

$\tau = \frac{l_t}{\lambda}$ - air surplus coefficient

l_t - theoretical fuel consumption need

Combustion efficiency coefficient is dependent on combustion chamber construction, for cal-

ulation purposes $\zeta_{CC} = 0.95$.

Pressure loss in combustion chamber consists of thermal and hydraulic losses. Considered as constant:

$$\sigma_{CC} = \sigma_t \sigma_h = 0.95 \quad (2.21)$$

From equations (2.20, 2.21) total temperature and pressure at combustion chamber outlet can be calculated:

$$T_3^* = \frac{i_3^*}{c_p'} \quad (2.22)$$

$$p_3^* = p_2^* \sigma_{CC} \quad (2.23)$$

2.5 Turbine

The turbine transforms energy from the hot exhaust gases to mechanical energy to drive the compressor. Similar to the compressor, the turbine model relies on performance maps describing the flow characteristics and efficiency. However, for a single-spool turbojet engine used in cruise missiles, the turbine typically operates in a choked flow regime for most of the operational envelope.

As illustrated in the generated turbine map Fig.10, the corrected mass flow ($\dot{m}_{t,corr}$) increases with the turbine pressure ratio (π_t) only up to a critical value. Beyond this point, the flow becomes choked, and the corrected mass flow remains constant, independent of the downstream pressure. This characteristic is modeled as:

$$\dot{m}_{t,corr} = f(\pi_t, n_{corr}) \quad (2.24)$$

The efficiency of the turbine (η_t), shown in Fig.11, is primarily a function of the pressure ratio and rotational speed (often correlated with the blade-to-gas speed ratio u/c).

Similarly to obtain physical values for turbine component, the dimensionless map outputs are scaled by the design nominal parameters (nominal exhaust gas flow $\dot{m}'_{nom}$, nominal turbine efficiency $\eta_{t,nom}$):

$$\begin{aligned} \dot{m}'_{corr} &= \dot{m}'_{map} \cdot \dot{m}'_{nom} \\ \eta_t &= \eta_{t,map} \cdot \eta_{t,nom} \end{aligned} \quad (2.25)$$

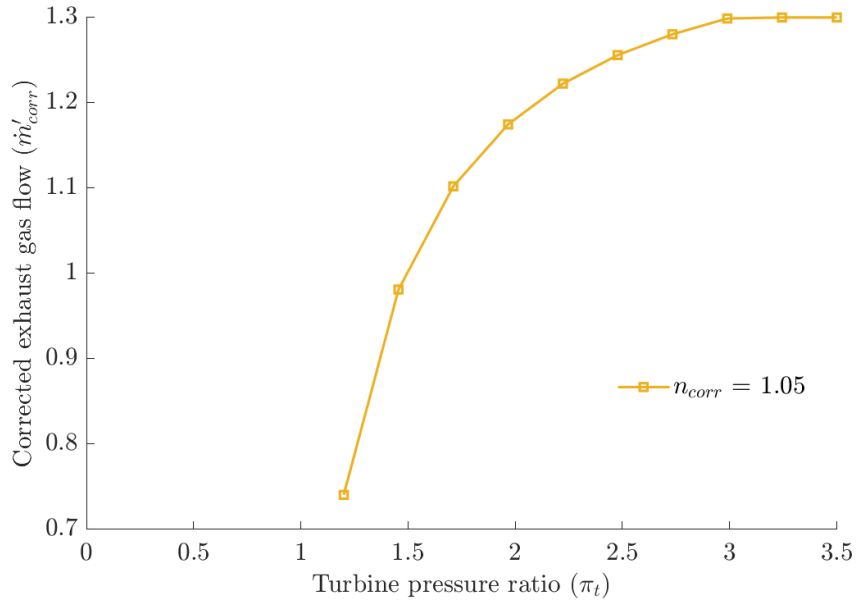


Figure 10: Turbine flow characteristic showing the choked flow regime.

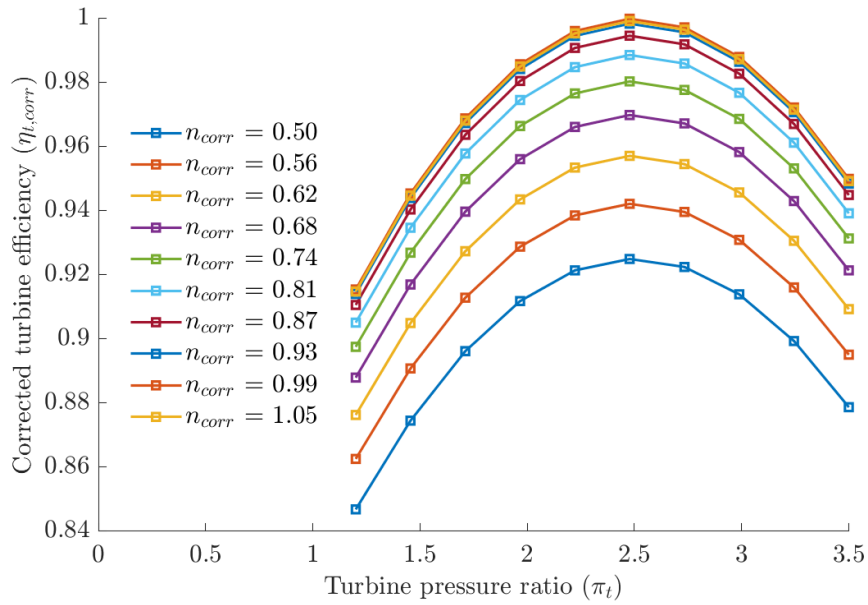


Figure 11: Turbine isentropic efficiency map.

Likewise compressor, turbine characteristics in Matlab/SIMULINK environment compressor characteristics are represented as 2D-Lookup Table. and are presented as 3D surfaces on figures (Fig.12, Fig.13).

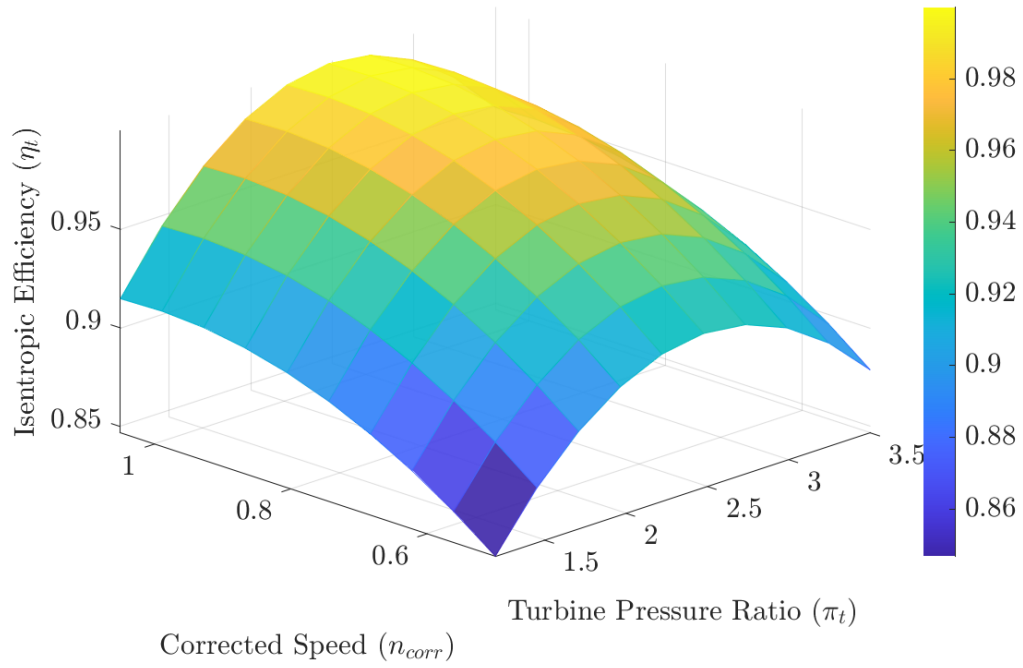


Figure 12: 3D surface of turbine corrected efficiency characteristic.

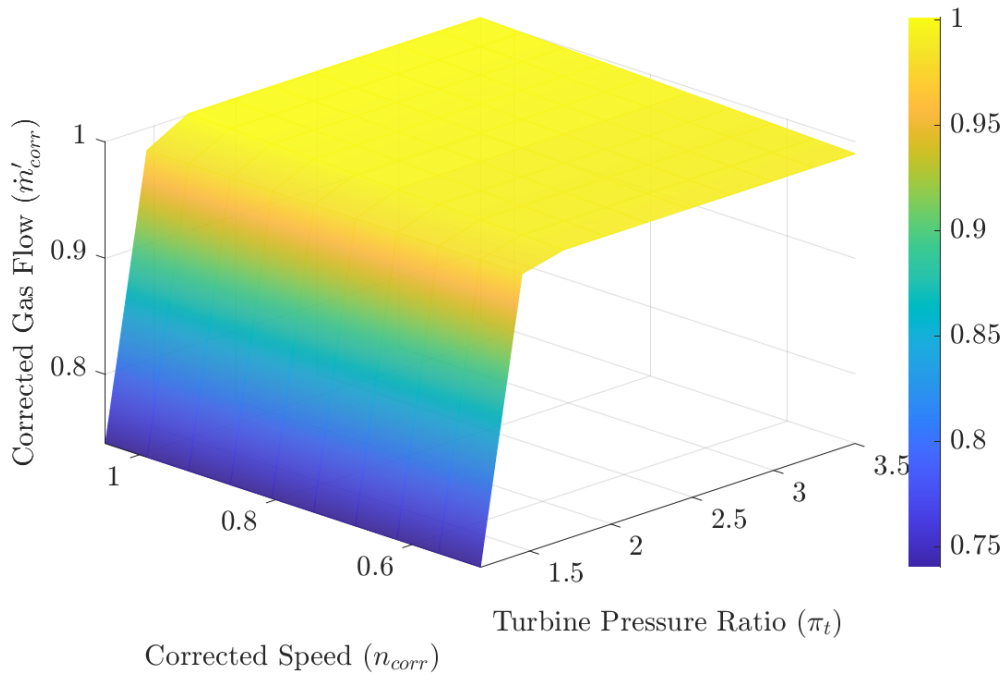


Figure 13: 3D surface of turbine corrected exhaust gas flow characteristic.

The temperature drop across the turbine is determined by the expansion work:

$$T_6^* = T_3^* \left(1 - \eta_t \left(1 - \pi_t^{\frac{1-\kappa'}{\kappa'}} \right) \right) \quad (2.26)$$

where:

$\pi_t = p_3^*/p_4^*$ – turbine pressure ratio.

By combining equations (2.22, 2.26) power of turbine can be calculated:

$$P_T = \dot{m}_t c_p' (T_3^* - T_4^*) \eta_m \quad (2.27)$$

where:

η_m – mechanical efficiency of the shaft.

2.6 Nozzle

Last part of jet engine with purpose of conversion fraction of exhaust gas enthalpy to kinetic energy while mass of fuel is negligible. Additional energy speeds up air flow at nozzle exit which leads to thrust generation.

$$K = \dot{m}'(c_8 - c_H) + A_8(p_5 - p_H) \quad (2.28)$$

In jet engines converging nozzle has to be considered, exhaust gas flow is decompressed fully ($p_8 = p_H$) when decompression rate is bellow critical value ($\pi_n < \pi_{cr}$).

$$\pi_n = \frac{p_6^*}{p_H} \quad (2.29)$$

$$\pi_{cr} = \left(\frac{\kappa' + 1}{2} \right)^{\frac{\kappa'}{\kappa' - 1}} \quad (2.30)$$

Equation of exhaust gas speed at nozzle exit is dependent on decompression type if ($\pi_n < \pi_{cr}$):

$$c_8 = \varphi_n \sqrt{2 \frac{\kappa'}{\kappa' - 1} R' T_6^* \left(1 - \frac{1}{\pi_n^{\frac{\kappa' - 1}{\kappa'}}} \right)} \quad (2.31)$$

In case on not compleat decompression ($\pi_n > \pi_{cr}$).

$$c_8 = \varphi_n \sqrt{2 \frac{\kappa'}{\kappa' + 1} R' T_6^*} \quad (2.32)$$

Also pressure needs to calculated.

$$p_8 = \frac{p_6^*}{\pi_{cr}} = p_6^* \left(\frac{2}{\kappa' + 1} \right)^{\frac{\kappa'}{\kappa' - 1}} \quad (2.33)$$

2.7 Shaft dynamics

Shaft dynamics is crucial to compute in model as spool rotational speed is needed for turbine and compressor characteristics. It is achieved from calculation of turbine and compressor torque.

$$J\dot{\omega} = T_T - T_C \quad (2.34)$$

Which are derived from power equations (2.19, 2.27) and rotational and angular speed dependency.

$$\omega = \frac{2\pi n}{60} \quad (2.35)$$

$$T_T = \frac{60P_T}{2\pi n} \quad (2.36)$$

$$T_C = \frac{60P_C}{2\pi n} \quad (2.37)$$

By substituting equation (2.34) with equations (2.35, 2.36, 2.37) shaft rotational speed differential equation can be derived.

$$\dot{n} = \frac{900}{J\pi^2 n} (P_T - P_C) \quad (2.38)$$

For computational model purposes rotational speed is taken from integration of equation (2.38).

2.8 Model implementation

The mathematical model based on equations derived in preceding sections has been implemented within Matlab/Simulink environment. The simulation adopts a modular Component Level Model (CLM) architecture, where each major engine component (inlet, compressor, combustion chamber, turbine, nozzle) is a separate subsystem. This approach allows for independent verification of each module, and gives opportunity for model scalability.

The signals flow within model mimics the real path of the air flow through the engine. Thermodynamic parameters are passed downstream from one block to the next. Dynamics and air flow variability is facilitated by two feedback loops.

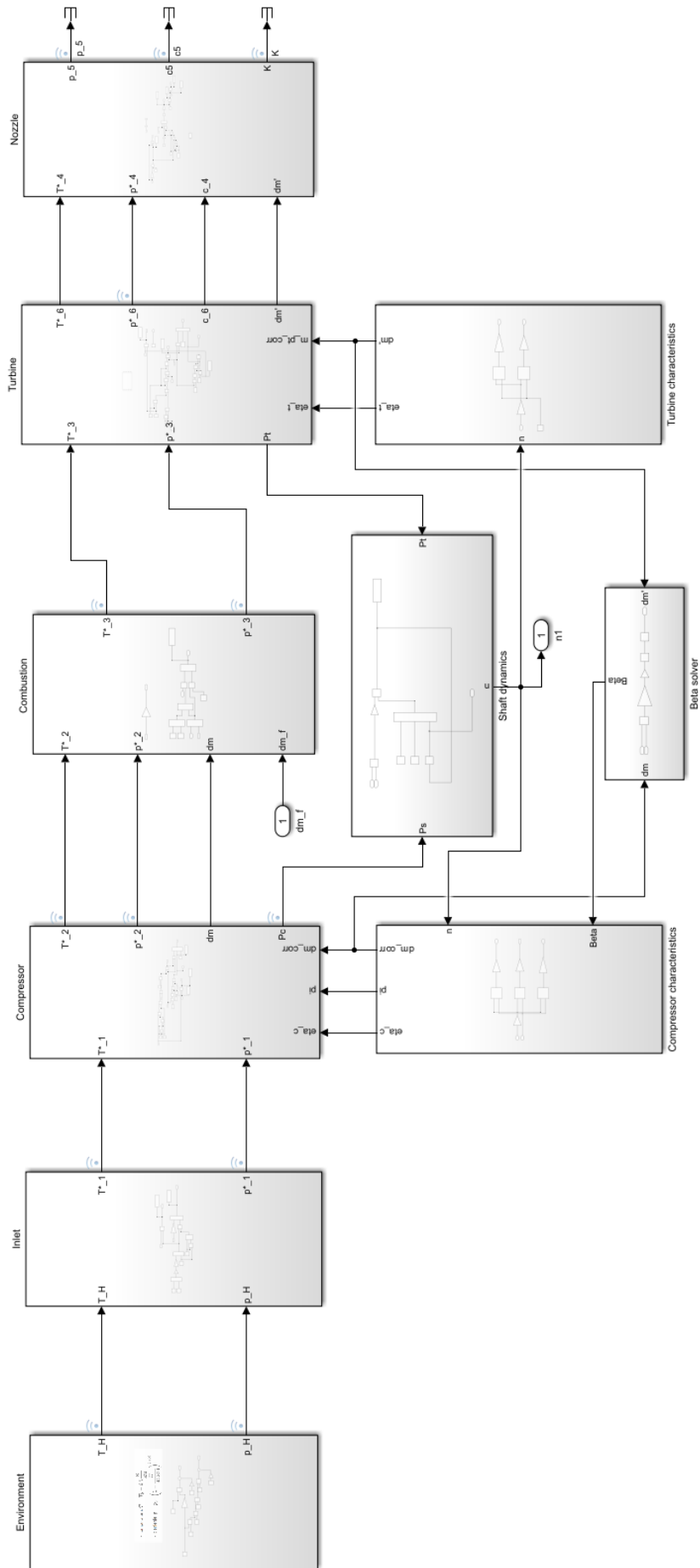


Figure 14: Developed turbojet engine reference model implemented in MATLAB/Simulink.

Mechanical loop, represented by shaft dynamics connects compressor and turbine by torque imbalance to calculate the rotational speed, which is then fed back to thermodynamical blocks.

Flow continuity loop, represented by beta solver block implements the iterative solution described in (2.14). It compares the compressor airflow with the flow reference from the turbine and adjust the β value using an integral to satisfy mass conservation law.

The performance characteristics of the engine are calculated from the thermodynamic equations and are located in dedicated blocks ("Compressor characteristics", "Turbine characteristics"). These blocks utilize 2-D lookup tables based on the maps presented in Fig.7 - Fig.9 and Fig.12 - Fig.13. The complete block diagram of the developed non-linear reference model is presented in Fig. 14.

3 Model linearization

Linearization is performed in order to be able to develop predictive control algorithm, which will be proposed in next chapter. For linearization defining model states (x) and inputs (u) and outputs (y) is needed to represent linear model as state-space equations.

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}\tag{3.1}$$

where:

A — state matrix,

B — input matrix,

C — output matrix,

D — disturbance matrix.

Unlike [4] and [13], different state vector has been proposed.

$$\mathbf{x} = \begin{bmatrix} n \\ \beta \end{bmatrix}\tag{3.2}$$

Purpose of this approach is to provide controllability of compressor reserve of stability, hence avoiding possibly destructive surge. Due to high maneuverability of missile, jet engine is more prone to enter instability. Output of the model is rotational speed (n) and input is fuel flow.

$$\begin{aligned}y &= n \\ u &= \dot{m}_f\end{aligned}\tag{3.3}$$

3.1 Finding steady state around operating point

Linearization can be performed in operating point. Linearized model state has to be close to operating point around which linearization has been performed due to rising discrepancy with reference model. For the purpose of finding steady state operating point has to be chosen. In scope of control loop which will be proposed in next chapter nominal rotational speed of the engine shaft has been chosen. In case of object considered in this thesis ($n_{nom} = 16000\text{RPM}$).

For finding steady state trimming method has been used. This process is commonly used in control systems for complex, nonlinear systems such as jet engine. The trimming process iteratively finds input value where state derivative tends to zero by using minimization algorithms like gradient descent.

3.2 Numerical linearization method

Linearization of nonlinear model has been performed by usage of Model Linearizer tool which is part of Simulink Control Design Toolbox. Mathematics behind operations executed by Simulink environment is expansion of nonlinear state functions to Taylor series, omitting high order components. Based on derived series algorithm calculates Jacobian matrix. Simulink blocks have predefined Jacobians that are then connected to chains in order to derive global linear model. Turbine and compressor are linearized by numerical perturbation. This method is based on inserting small perturbation (ϵ) in order to calculate state function partial derivative based on measured output derivative.

$$\frac{\partial f}{\partial x_i} = \frac{f(x + \epsilon, u) - f(x, u)}{\epsilon} \quad (3.4)$$

Obtained model is linear and time-invariant, that will be base for control loop. Linearized model can be represented with state-space equation (3.1) and numerically derived A,B,C,D:

Numerically derived state space matrices A,B,C,D:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} -0.2244 & 1764 \\ -0.00067 & -4.05 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 8365 \\ 0 \end{bmatrix} \\ \mathbf{C} &= \begin{bmatrix} 8365 \\ 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned} \quad (3.5)$$

Obtained model is linear and time-invariant, that will be base for control loop. Linearized model can be represented with state-space equation (3.1) and algebraic representation of (3.5):

$$\begin{aligned} \begin{bmatrix} \dot{n} \\ \dot{\beta} \end{bmatrix} &= \begin{bmatrix} -0.2244 & 1764 \\ -0.00067 & -4.05 \end{bmatrix} \begin{bmatrix} n \\ \beta \end{bmatrix} + \begin{bmatrix} 8365 \\ 0 \end{bmatrix} \dot{m}_f \\ n &= \begin{bmatrix} 8365 \\ 0 \end{bmatrix} \begin{bmatrix} n \\ \beta \end{bmatrix} \end{aligned} \quad (3.6)$$

4 Control loop

During flight, the engine shaft rotates at a nominal axial speed (n_{nom}). However, the missile's maneuvers will impact steady-state operation. These maneuvers alter the inlet air flow parameters, affecting each stage of the engine. Consequently, this disturbance must be corrected by adjusting the control input, which is the fuel flow ($\dot{m}_f$). To achieve this optimally, an appropriate controller and feedback loop must be introduced.

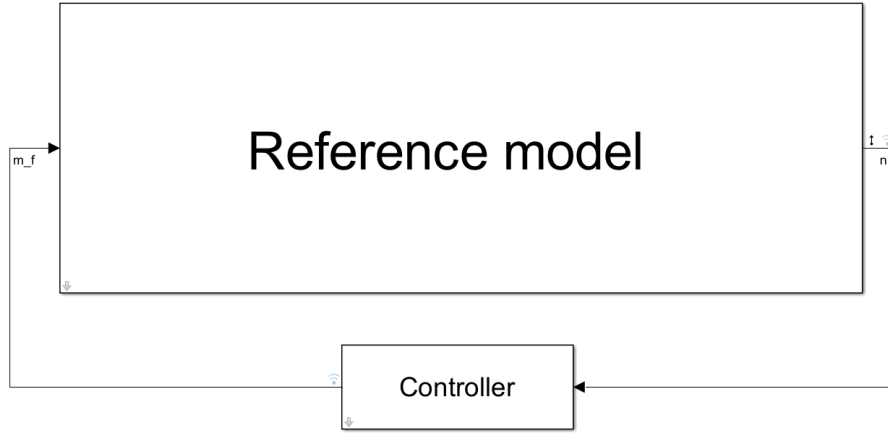


Figure 15: Block diagram of the reference model and feedback controller.

4.1 Controlled object

The controlled object is the reference model described in Chapter 3, developed in the MATLAB Simulink environment shown on Fig.14. The controller is based on a feedback loop. The input to the controller is the rotational speed, and the output is the control signal, fuel flow [7].

4.2 PI control

The Proportional-Integral (PI) controller has been an industry standard in control theory for decades. The working principle of the PI controller is to minimize the error (e) by adjusting the controller output, which influences the model state. Here, e is defined as the difference between the desired setpoint (S_{ref}) and the actual measured value (S_m).

$$e(t) = S_{ref} - S_m \quad (4.1)$$

PI control function.

$$u(t) = K_p e(t) + K_I \int e(t) dt \quad (4.2)$$

The proportional part multiplies the error by the proportional gain (K_P) and the integral part minimizes the error over time. The control output can be tuned by tweaking K_P and the integral gain (K_I). Higher gains values leads to faster regulation response but can lead to overshoot, instability and longer settling time. In the other side of the spectrum to low gains will not provide fast enough response. Proportional Integral control is a subset of Proportional Integral Derivative (PID) controller. Derivative part has been skipped intentionally due to noisy

rotational speed measurement generated by shaft vibrations. Aggressive filters would have to be applied what could lead to slower regulator response to fast changes in axial velocity which have to be considered in missile engine with lower moment of inertia and higher dynamic.

Drawbacks of PI control are unawareness of plant physical limitation for example rate of rotational speed change, delay of fuel pump actuators or maximal fuel flow. That could lead to controller saturation, in this case integral part of control function is summing up to minimize error which rate of change is limited by outside factors. Anti-windup is used to prevent this situation. Two methods are widely used clamping and back-calculation. First method clamps integrator when output is saturated, second feeds back difference between real and saturated output to reduce the integrated value. This method helps to return from saturation faster.

For the purpose of this thesis PI controller with back calculation anti-windup in Matlab/Simulink has been implemented shown in Fig.16.

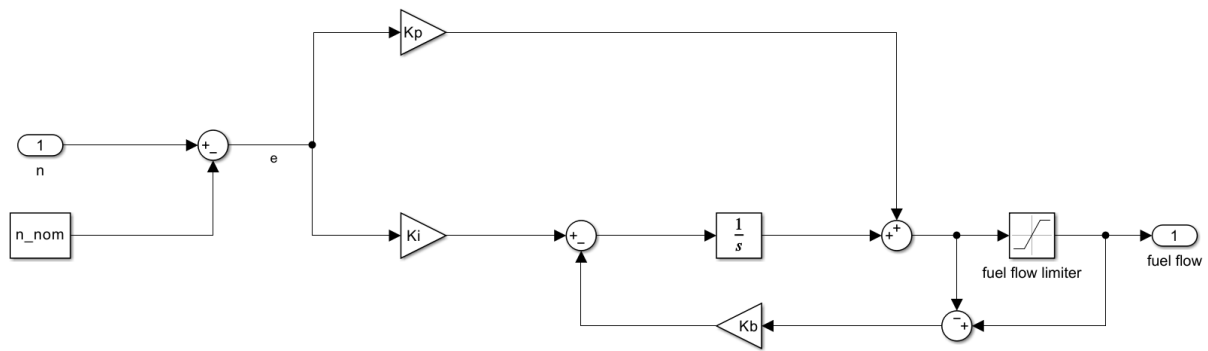


Figure 16: PI controller with back calculation anti-windup

Values of controller gains have been presented on Table 3.

Table 3: PI controller gains values

Variable name	Value	Unit
proportional gain K_p	0.01	-
integral gain K_I	0.005	-
back calculation gain K_b	5	-
minimal fuel flow $m_{f,min}$	0	kg/s
maximal fuel flow $m_{f,max}$	0.1	kg/s

4.3 Predictive control

Historically, Model Predictive Control was primarily applied to slow-dynamic process industries (e.g., chemical or petrochemical plants) due to the significant computational burden required to solve optimization problems. However, with the continuous advancement in microprocessor

technology and the increasing processing power of modern embedded systems, MPC has become a viable and increasingly popular solution for fast-dynamic systems, including aerospace propulsion. The ability of contemporary microcontrollers to perform complex matrix operations in real-time allows for the implementation of advanced optimization algorithms directly on the full authority digital engine control (FADEC) [2] [10].

While the PI controller described in the previous section relies on past errors to determine the control input, Model Predictive Control (MPC) represents a more advanced control strategy that utilizes an explicit mathematical model of the plant to predict the future behavior of the system. In the context of turbojet engine control, where safety margins and operational limits are critical, MPC offers significant advantages over classical feedback loops. The fundamental principle of MPC is based on the receding horizon strategy. At each sampling instant, the controller solves a constrained optimization problem over a finite prediction horizon (N_p). This optimization aims to find a sequence of future control increments (fuel flow adjustments) that minimizes a specific cost function while satisfying system constraints. Only the first element of the calculated control sequence is applied to the engine, and the entire optimization process is repeated at the next time step with updated state measurements. The primary motivation for implementing MPC in cruise missile propulsion is its inherent ability to handle constraints explicitly. A jet engine operates under strict physical limitations. Respecting the physical range and rate of change of the fuel metering unit (m_f). Unlike PI control, which requires anti-windup schemes or separate from controller limiters to handle these constraints, MPC incorporates them directly into the optimization problem.

The discrete-time state-space model of the engine can be described as:

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) \end{aligned} \quad (4.3)$$

By iterating this model over the prediction horizon N_p , the future state variables can be expressed as a function of the current state $x(k)$ and future control inputs $u(k+i)$. The prediction vector for the output $\mathbf{Y}$ is formulated as:

$$\mathbf{Y} = Fx(k) + \Phi = \mathbf{U} \quad (4.4)$$

where the vectors are defined as:

$$\mathbf{Y} = \begin{bmatrix} y(k+1) \\ y(k+2) \\ \vdots \\ y(k+N_p) \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} u(k) \\ u(k+1) \\ \vdots \\ u(k+N_p-1) \end{bmatrix} \quad (4.5)$$

And the matrices F and Φ are derived from the system matrices (A, B, C) :

$$F = \begin{bmatrix} CA \\ CA^2 \\ \vdots \\ CA^{N_p} \end{bmatrix}, \quad \Phi = \begin{bmatrix} CB & 0 & \cdots & 0 \\ CAB & CB & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{N_p-1}B & CA^{N_p-2}B & \cdots & CB \end{bmatrix} \quad (4.6)$$

To compute the optimal control sequence, a quadratic cost function J is minimized. The objective is to minimize the error between the predicted output and the reference trajectory while penalizing excessive control efforts:

$$J = \sum_{i=1}^{N_p} \|y(k+i) - r(k+i)\|_Q^2 + \sum_{i=0}^{N_u-1} \|\Delta u(k+i)\|_R^2 \quad (4.7)$$

where:

- N_p is the prediction horizon,
- N_u is the control horizon,
- $r(k+i)$ is the reference signal (desired rotational speed),
- $\Delta u(k+i)$ represents the change in control input (fuel flow increment),
- Q and R are weighting matrices determining the trade-off between tracking accuracy and control smoothness.

The optimization problem is solved at each step subject to constraints:

$$\begin{aligned} u_{min} &\leq u(k+i) \leq u_{max} \\ \Delta u_{min} &\leq \Delta u(k+i) \leq \Delta u_{max} \\ y_{min} &\leq y(k+i) \leq y_{max} \end{aligned} \quad (4.8)$$

Solving the optimization problem (4.4) yields a sequence of optimal control increments:

$$\Delta \mathbf{U}^* = [\Delta u^*(k), \Delta u^*(k+1), \dots, \Delta u^*(k+N_u-1)]^T \quad (4.9)$$

According to the receding horizon strategy, only the first element of this sequence is applied to the plant at the current time step k :

$$u(k) = u(k-1) + \Delta u^*(k) \quad (4.10)$$

The remaining calculated values are discarded, and the optimization is repeated at step $k+1$.

Furthermore, a distinction is made between the prediction horizon (N_p) and the control horizon (N_u), where typically $N_u < N_p$. It is assumed that control adjustments are only made up to the control horizon. Beyond this point, the control increments are zero:

$$\Delta u(k+i) = 0 \quad \text{for} \quad N_u \leq i < N_p \quad (4.11)$$

This implies that the control input remains constant for the remainder of the prediction window, maintaining the value of the last calculated step:

$$u(k+i) = u(k+N_u-1) \quad \text{for} \quad i \geq N_u \quad (4.12)$$

This assumption reduces the computational complexity of the optimization problem by limiting the number of decision variables to N_u .

In order to mitigate steady state error associated with MPC due to imperfection of linearization process, integral part will be added, hence state of equilibrium error will be reduced without compromising accurate transient response. To achieve this setpoint of the regulator will be adjusted by disturbance calculated from difference between measured and predicted output:

$$d = n_{measured} - n_{predicted} \quad (4.13)$$

Disturbance (d) is additionally integrated and multiplied by gain α to control speed of error correction and ensure stability during fast setpoint change:

$$d_{filtered} = \int (\alpha d) dt \quad (4.14)$$

For deeper understating and customization possibility MPC has been implemented in C++ programming language [6]. Matrix calculations were performed with help of Eigen library. Most important source code parts are presented on Listings 1.

```

1 void ModelPredictiveController::updateState(MatrixXd current_x) {
2     currentState = current_x;
3
4     // Initialization
5     if (isFirstRun) {
6         internalStatePrediction = current_x;
7         isFirstRun = false;
8     }
9
10    // Measured output
11    MatrixXd y_meas = C * currentState;
12
13    // Predicted output
14    MatrixXd y_pred = C * internalStatePrediction;
15
16    // Disturbance calculation
17    MatrixXd raw_disturbance = y_meas - y_pred;
18
19    // Disturbance filtration
20    //  $d(k) = (1-\alpha)d(k-1) + \alpha \cdot \text{raw\_error}$ 
21    estimatedDisturbance = (1.0 - alphaFilter) * estimatedDisturbance
22                          + alphaFilter * raw_disturbance;
23
24    // Setpoint correction
25    MatrixXd vectorS = desiredTrajectoryHorizon - phi * currentState;
26
27    // Input in control horizon calculation
28    MatrixXd inputSequence = gainMatrix * vectorS;
29    computedInput = inputSequence.block(0, 0, m, 1);
30
31    // State prediction update
32    internalStatePrediction = A * currentState + B * computedInput;
33 }

```

Listing 1: Disturbance estimation and controller output and state prediction

5 Simulation scenarios

The verification of the proposed control strategies has been conducted through a series of simulations representing critical phases of engine operation and external disturbances. The analysis focuses on comparing the performance of two controller configurations: classical PI and Model Predictive Control (MPC). The initial drop in rotational speed observed in the plots results from the initialization of the fuel flow state and should be disregarded in the context of the controller performance analysis.

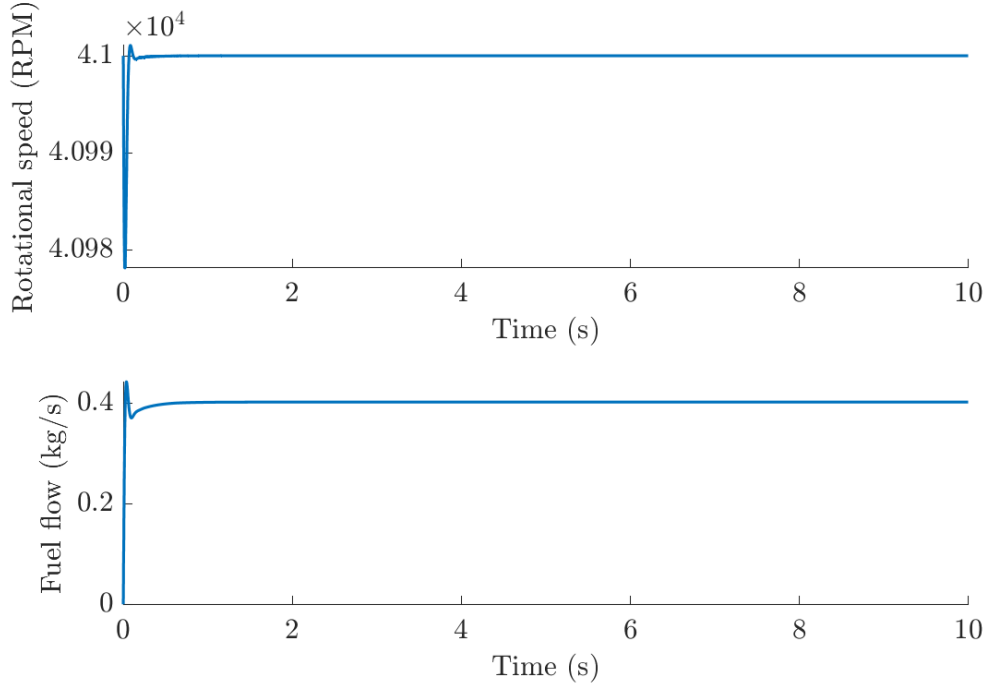


Figure 17: Steady state at nominal speed

5.1 Ramp up

This scenario simulates a command for rapid acceleration, typical of a takeoff run or a sudden demand for maximum thrust. The reference rotational speed (n_{set}) is increased from an idle state through cruise to a nominal speed.

As illustrated in Fig.18 and Fig.19, the initial rise time in the first phase of acceleration is identical for both controllers. This indicates that the system is operating under saturation—the requested fuel flow reaches the physical limit of the fuel pump (clamped at maximum value). In this phase, the engine's inertia is the limiting factor, and no control algorithm can overcome this physical constraint.

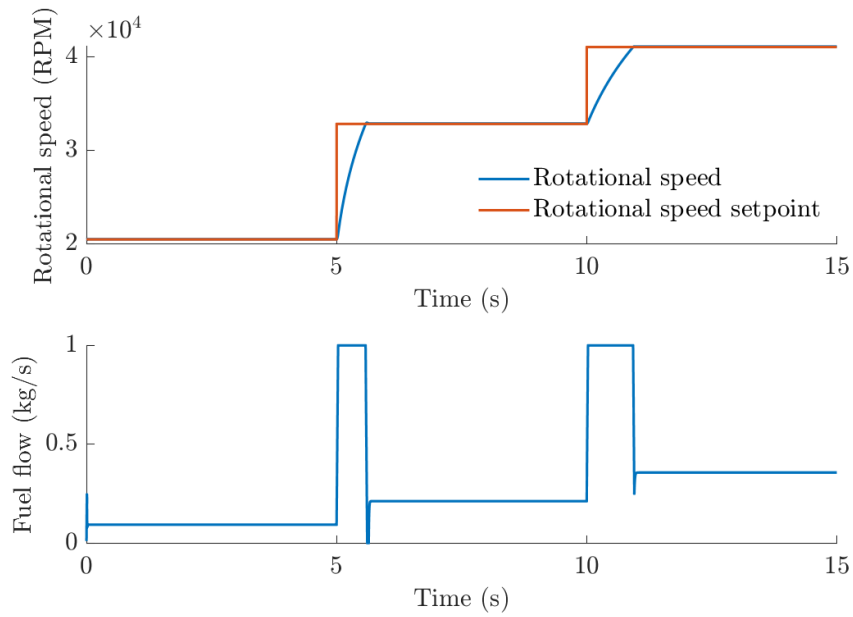


Figure 18: Two stage ramp up controlled by MPC

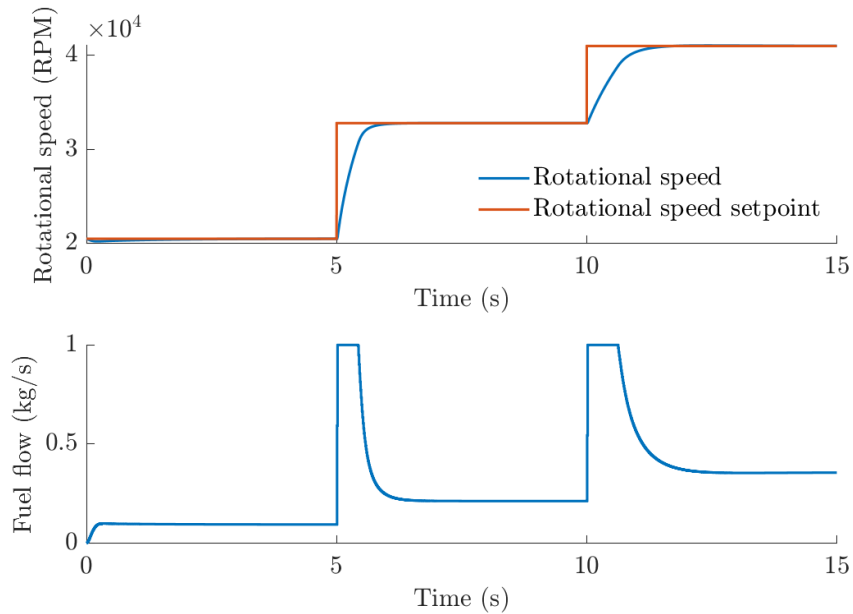


Figure 19: Two stage ramp up controlled by PI

However, a significant difference is observed in the approach to the setpoint. The MPC algorithm, leveraging its prediction horizon, anticipates the target speed and reduces the fuel flow *before* the setpoint is reached. This results in a "soft landing" on the target value with minimal settling time. In contrast, the PI controller, which reacts only to the current error, maintains high fuel flow for too long, leading to integral windup (despite anti-windup measures) or a slower settling phase as seen in Fig.19.

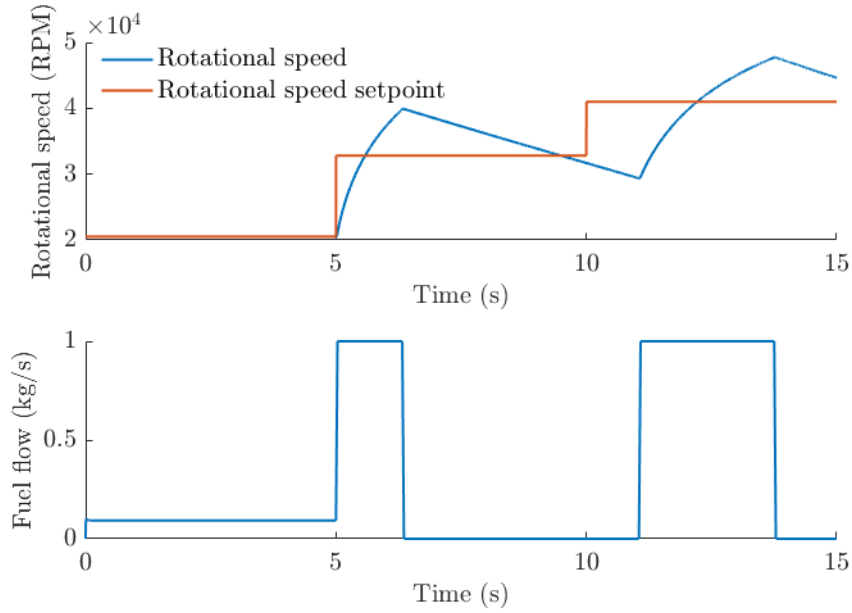


Figure 20: Two stage ramp up controlled by PI without anti-windup

Fig.20 highlights the critical importance of the back-calculation anti-windup method implementation. Without it, the integral term accumulates excessive error during the saturation phase, leading to significant overshoot and potentially dangerous oscillations.

5.2 Rapid descent

This scenario replicates a fast missile descent ($200 \frac{m}{s}$) and hovering at low altitude before finishing its mission. Theoretically, the engine RPM should remain constant at the nominal speed.

The importance of active speed control is visualized in Fig.21. During the descent, the ambient pressure and air density at the inlet increase significantly. This alters the power balance between the turbine and the compressor. Without fuel flow correction, the engine undergoes a spontaneous ramp up, where rotational velocity rises rapidly, which could result in mechanical failure or structural damage due to overspeed.

The MPC response, visible in Fig.23, maintains a steady rotational speed throughout the entire descent. By utilizing the disturbance estimator (offset-free formulation), the MPC identifies the model mismatch caused by changing flight conditions and aggressively reduces fuel flow to counteract the spool-up tendency. Conversely, the PI controller shown in Fig.22 demonstrates a reactive behavior. It requires a significant error (RPM increase) to build up before the integral term sufficiently reduces the fuel flow, leading to a noticeable deviation from the setpoint.

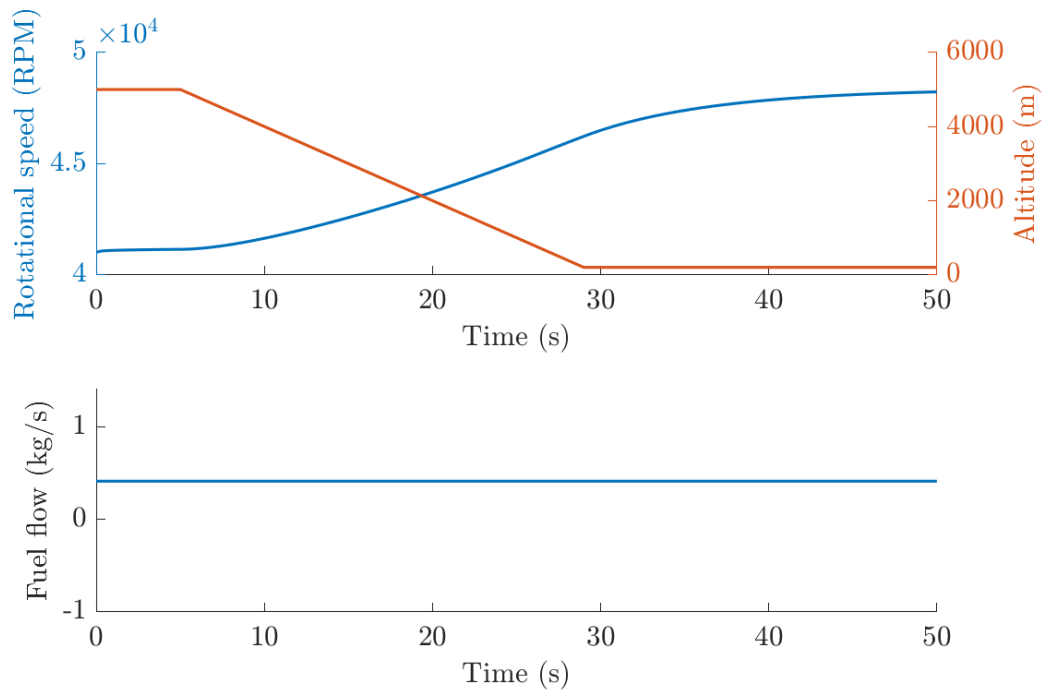


Figure 21: Rapid descent without control

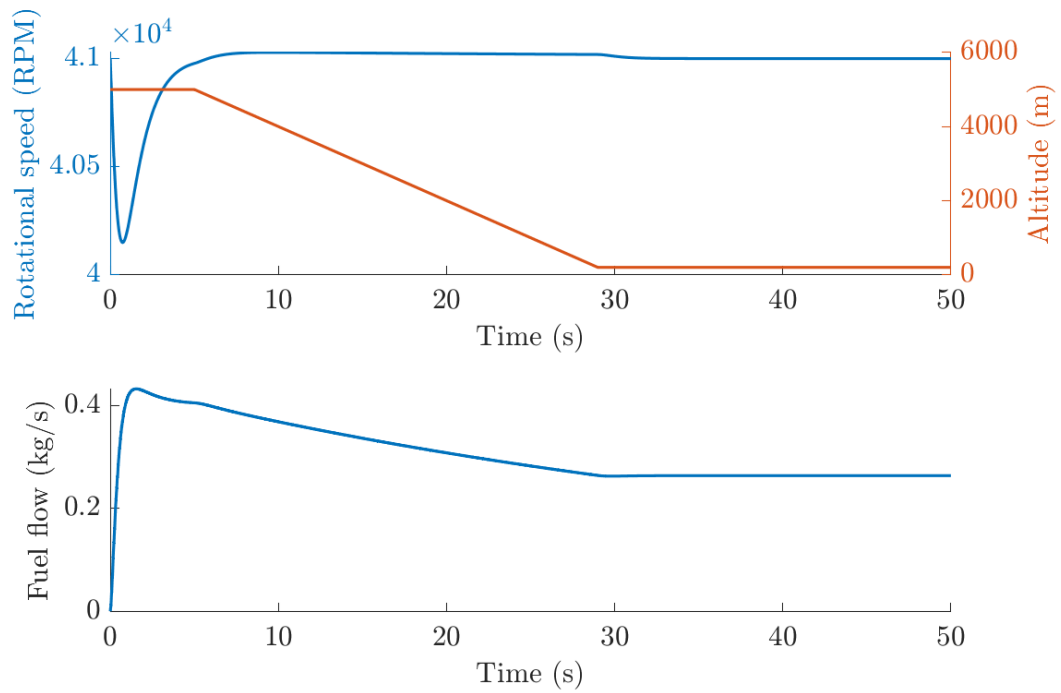


Figure 22: Rapid descent PI control

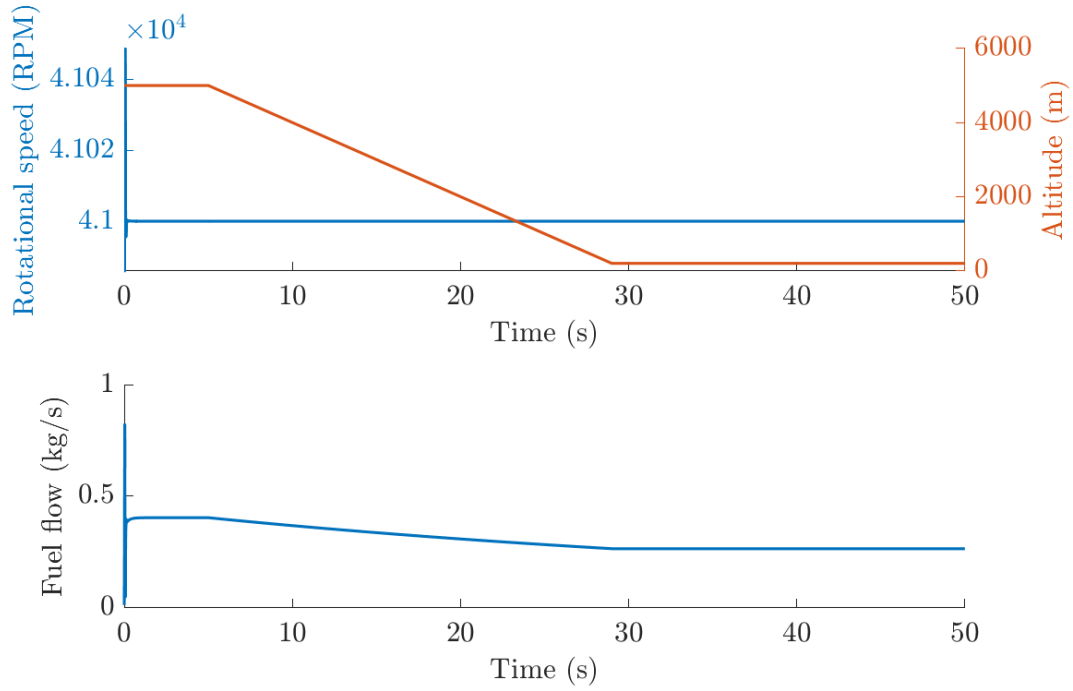


Figure 23: Rapid descent MPC

5.3 Exhaust gas suction

This scenario focuses on the ingestion of exhaust gas from a preceding engine or missile through the inlet to the compressor. This phenomenon leads to a change in the specific heat ratio (κ) and the gas constant, reducing the thermodynamic efficiency of the cycle and disturbing the power equilibrium.

As shown in Fig.24, maintaining constant fuel flow during this disturbance leads to a drastic drop in rotational speed (approximately 500 RPM in 1 second), which can threaten the engine's stability.

Reaction after applying control shown in Fig.25 and Fig.26. The comparative analysis in Fig.27 demonstrates that the MPC controller is significantly more effective in rejecting this disturbance. The MPC reacts with a rapid and decisive adjustment of the fuel flow (visible as a sharp peak in the control signal) to oppose the efficiency drop. The PI controller is significantly slower in its reaction, resulting in a deeper drop in RPM and a longer recovery time.

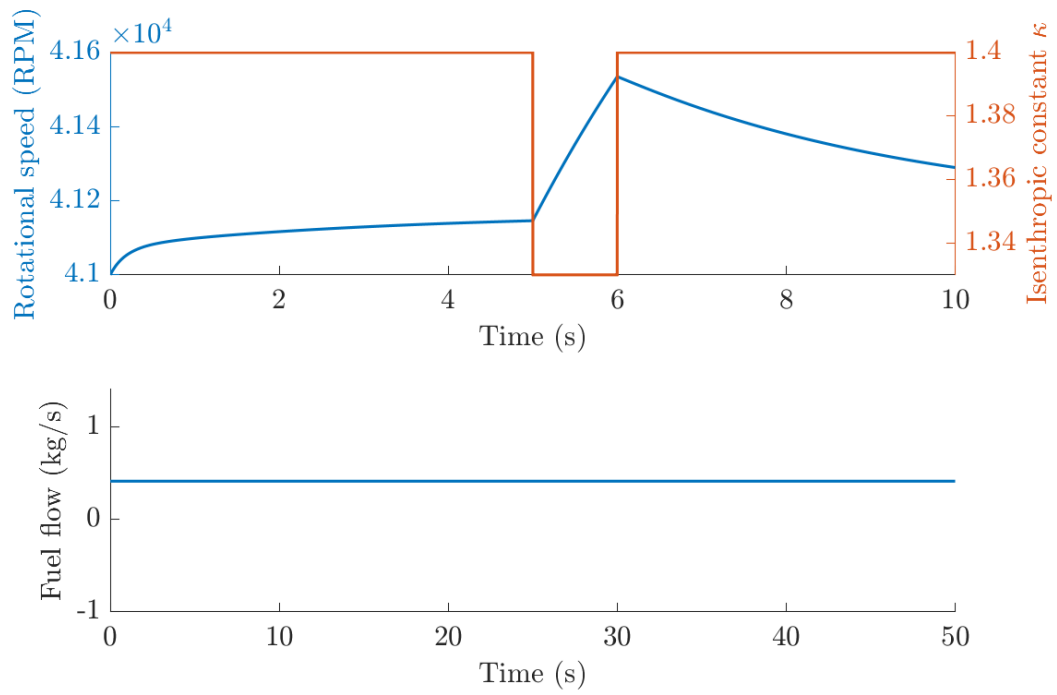


Figure 24: Exhaust gas suction without control

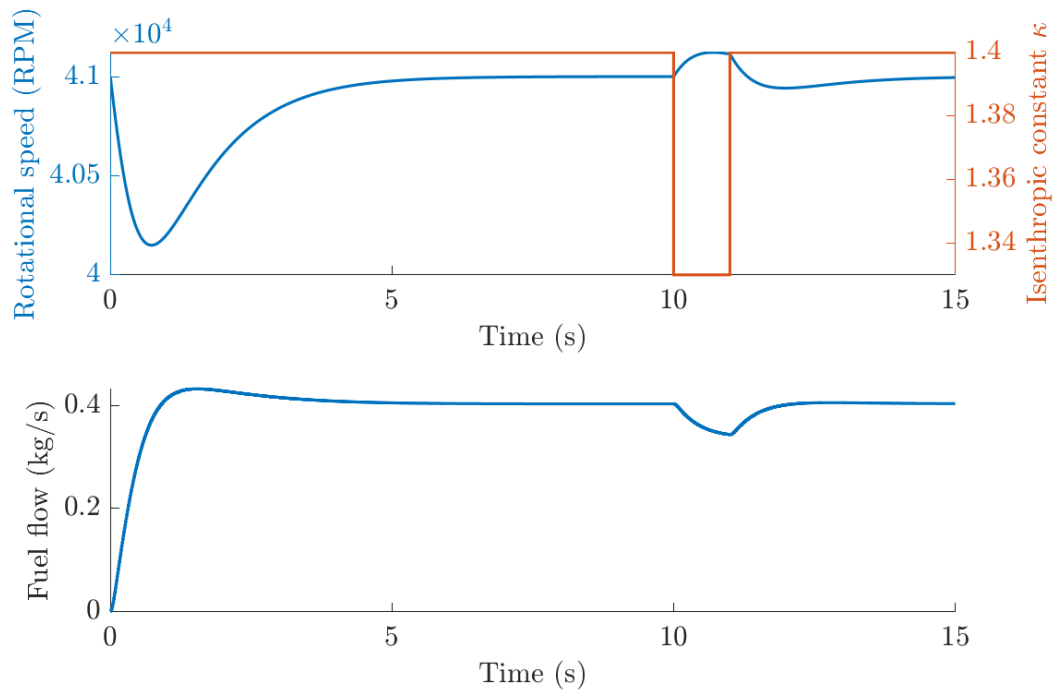


Figure 25: Exhaust gas suction with PI control

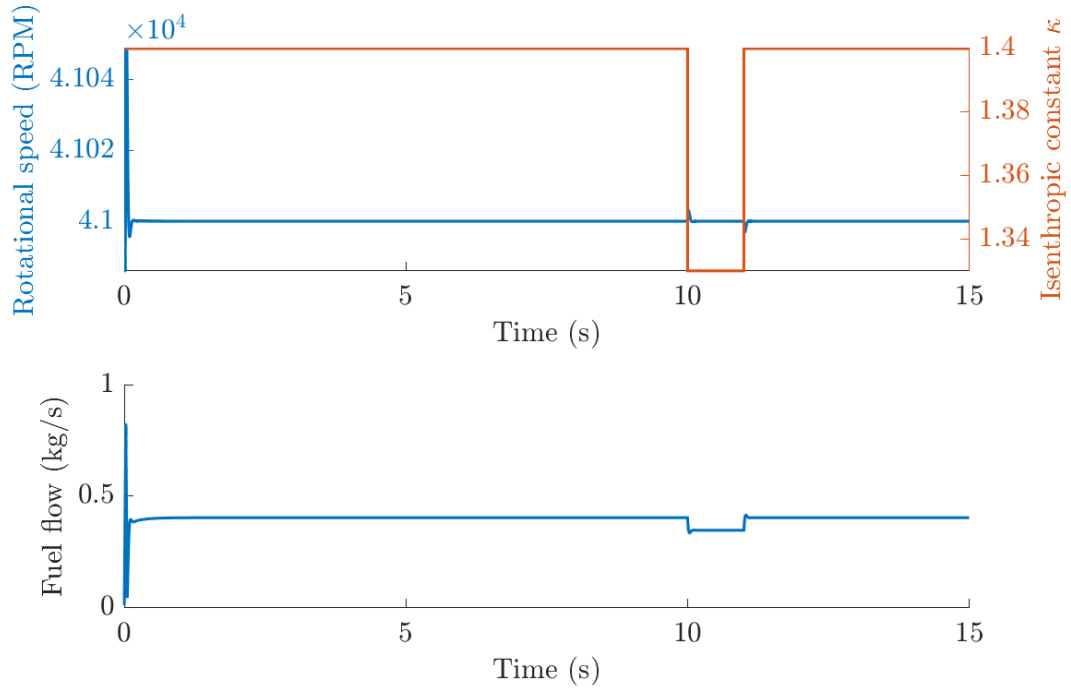


Figure 26: Exhaust gas suction with MPC control

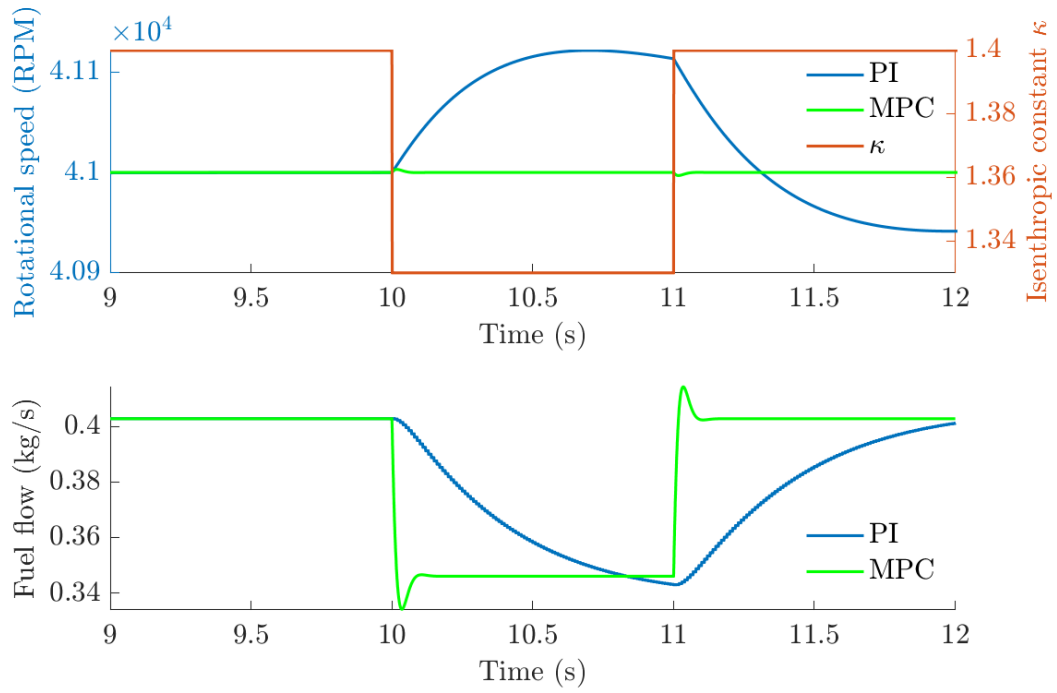


Figure 27: Exhaust gas suction MPC and PI close-up

5.4 Inlet pressure distortion

High maneuverability of a cruise missile may result in inlet pressure distortion (e.g., during high-angle-of-attack maneuvers). The consequences are physically similar to the aforementioned

exhaust gas suction, affecting the mass flow and pressure ratio. However, the simulated 10% pressure drop proved to be less severe than the thermodynamic change caused by κ variation.

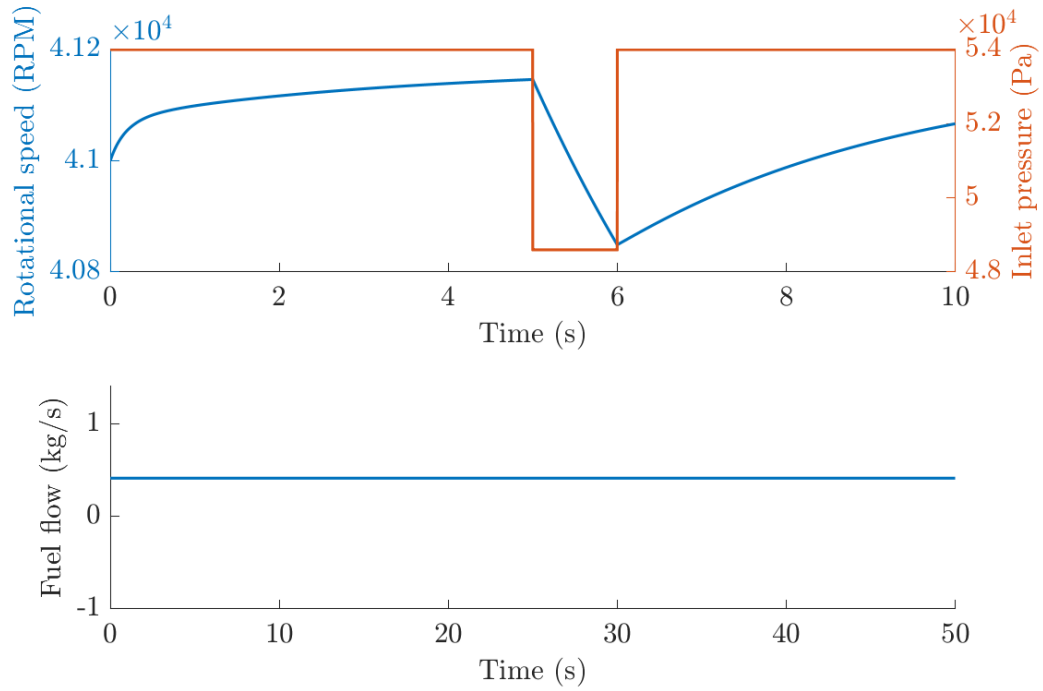


Figure 28: Inlet pressure distortion without control

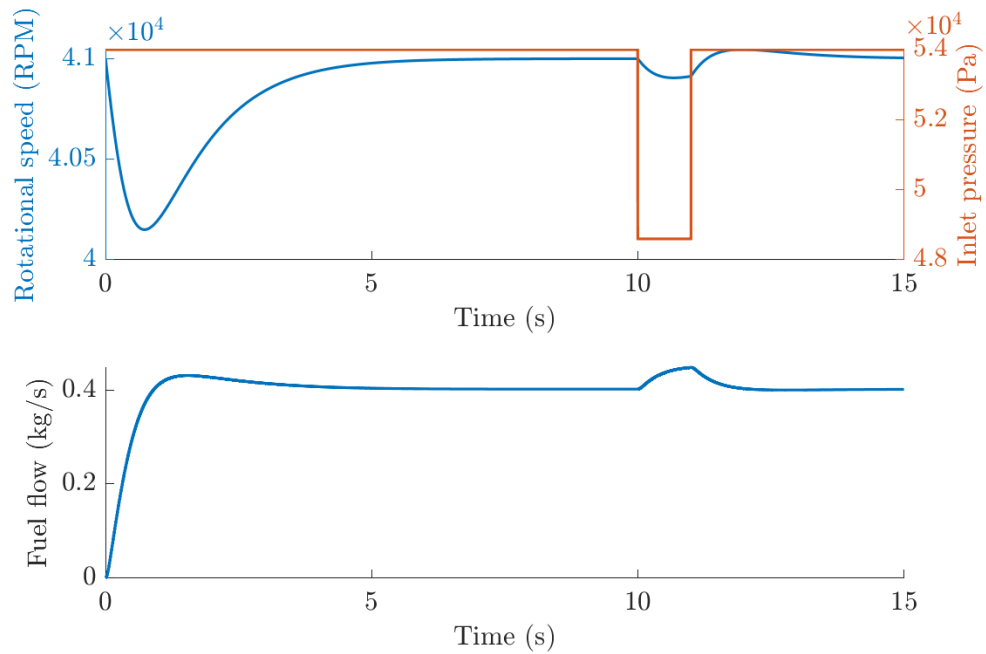


Figure 29: Inlet pressure distortion with PI control

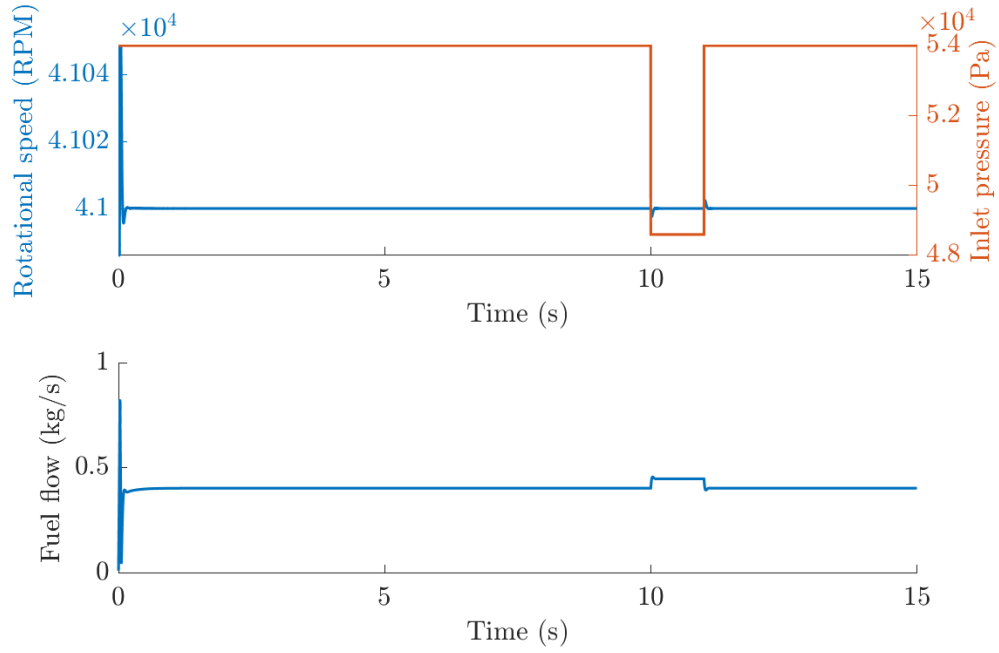


Figure 30: Inlet pressure distortion with MPC control

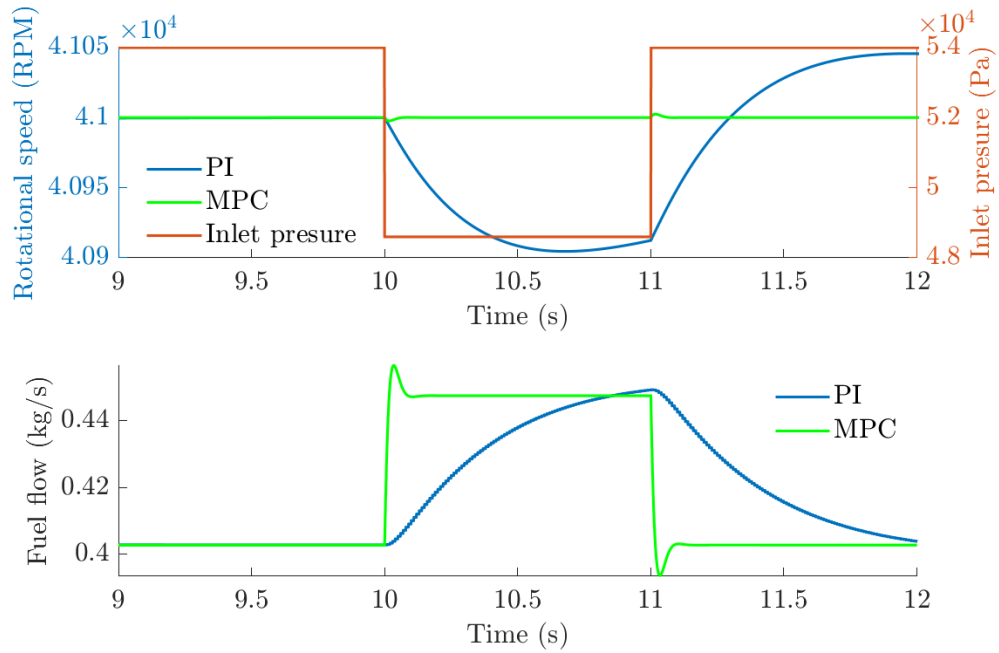


Figure 31: Inlet pressure distortion MPC and PI close-up

Object response to disturbance shown in Fig. 29 and Fig. 30. The performance comparison in Fig.31 confirms the conclusions drawn from previous scenarios. The MPC algorithm exhibits superior robustness, maintaining the engine operating point closer to the nominal value. This capability is crucial for preserving the surge margin during violent maneuvers, where pressure fluctuations are common. The stand-alone PI regulation allows for a larger dynamic error, which could potentially bring the compressor closer to the instability limit.

6 Conclusions

The research conducted in this thesis aimed to develop a numerical model of a turbojet engine and to design a modern control system capable of handling the dynamic constraints typical of cruise missile operation. The following conclusions can be drawn from the modeling process and the comparative analysis of the control strategies.

6.1 Evaluation of Control Strategies

The simulation results clearly demonstrate the superiority of the Model Predictive Control (MPC) algorithm over the classical Proportional-Integral (PI) controller in the context of high-performance propulsion systems.

The MPC algorithm successfully managed physical limitations, such as fuel flow saturation. By anticipating the setpoint approach, MPC eliminated the overshoot observed in the PI controller responses (visible in the ramp-up scenario). This predictive capability is crucial for extending the engine's lifespan by reducing thermal and mechanical stress caused by transient over-regulation.

The implementation of the offset-free MPC formulation (with a disturbance estimator) proved highly effective. In scenarios involving rapid descent and exhaust gas suction, the MPC maintained the rotational speed within a tight tolerance margin (deviation less than 0.1% in steady state), whereas the PI controller exhibited significant dynamic errors and slower recovery times. This confirms that MPC provides better robustness against unmeasured disturbances and model mismatches.

While the rise time in saturation-limited scenarios was identical for both controllers due to physical pump constraints, the MPC achieved a shorter settling time. This efficiency is critical for missile agility, allowing for faster thrust modulation during combat maneuvers. However, it must be noted that the performance of MPC relies heavily on the accuracy of the linear model. Significant deviations from the linearization point (n_{nom}) could potentially degrade control quality, although the implemented disturbance estimator successfully mitigated this risk in the tested scenarios.

Also, an engineering trade-off must be acknowledged. While the MPC provides superior precision, it imposes a significantly higher computational burden due to the necessity of solving a Quadratic Programming problem at every time step. In contrast, the classical PI regulator, despite its slower response and tendency for overshoot, remains a computationally lightweight and easily implementable solution. Therefore, for applications where hardware resources are limited and extreme precision or strict constraint handling is not the primary safety requirement, the PI controller remains a valid and effective option.

6.2 Evaluation of the Numerical Model

The developed Component Level Model (CLM) based on the "Beta-lines" method provided a stable and functionally accurate representation of a single-spool turbojet engine.

The adoption of a state vector $x = [n, \beta]^T$ instead of a simplified first-order inertial model allowed for a more realistic representation of compressor dynamics, including the monitoring of the surge margin. This level of detail is essential for developing safety-critical control laws.

The modular architecture of the simulation serves as a solid foundation for a "digital twin". The ability to simulate complex failure states (e.g., inlet distortion, gas ingestion) proves that the model is suitable not only for controller tuning but also for fault tolerance analysis.

Despite the increased complexity compared to basic transfer function models, the derived linearized state-space representation remained computationally efficient, enabling the real-time execution of the MPC optimization algorithm (solving the Quadratic Programming problem at each time step).

6.3 Final Remarks

The primary objective of this thesis, the development of a comprehensive numerical environment comprising a turbojet engine reference model and an advanced control system has been successfully achieved. The work demonstrates that it is possible to implement a complex, physics-based Component Level Model and a sophisticated Model Predictive Control algorithm within a unified simulation framework.

The comparative analysis with the classical PI regulator served as a validation tool for the developed system. It confirmed that the designed MPC controller is not only functional but also capable of operating correctly under dynamic conditions and constraints, offering a stable and high-performance solution for propulsion management. Predictive control however may not be the best choice for some

Consequently, the resulting software platform constitutes a verified engineering tool. It lays the groundwork for further development of autonomous aerial systems, enabling safe and cost-effective testing of propulsion integration with flight dynamics (6-DOF) or hardware-in-the-loop (HIL) applications.

7 Possible further areas of development

The modular Component Level Model (CLM) architecture adopted in this thesis ensures flexibility and facilitates future enhancements. While the current model provides a robust platform for control system design, several areas have been identified for potential development to increase physical fidelity and broaden the range of applications.

One significant enhancement would be the expansion of the exhaust system model. The current fixed-geometry nozzle could be upgraded to a variable exhaust nozzle with adjustable throat area (A_8). This modification is essential for modeling afterburning engines, as it allows for the management of backpressure during reheat operation. Furthermore, implementing thrust vectoring capabilities would transform the current Single-Input Single-Output (SISO) system into a complex multivariable control problem, relevant for modern high-maneuverability missiles [15].

To further bridge the gap between simulation and real-world physics, the thermodynamic model could be transitioned from a quasi-steady-state approach to a fully transient formulation. As proposed by Beneda [4], this involves introducing differential equations for pressure and temperature dynamics ($\frac{dp}{dt}, \frac{dT}{dt}$) within the component volumes (plenums). This modification would account for the mass and energy storage effects (volume packing and thermal inertia), thereby imposing realistic physical limits on the rates of change of thermodynamic parameters, which are instantaneous in algebraic models.

In terms of compressor fidelity, future iterations could incorporate varying blades angle and inter-stage bleed valves. These mechanisms are critical for improving the surge margin during rapid acceleration and deceleration, which are common operational scenarios for combat UAVs. Additionally, the current single-spool architecture serves as a baseline that can be extended to a multi-spool turbofan configuration [8]. By adding a low-pressure shaft and a bypass duct, the model could simulate high-bypass engines used in long-endurance drones or loitering munitions.

Finally, the developed propulsion model should not exist in isolation. A logical next step is the integration of the engine model into a larger 6-DOF (Six Degrees of Freedom) flight dynamics simulation of a missile or UAV. This would allow for the analysis of coupling effects, such as the impact of inlet distortion during high-G maneuvers on engine stability [16]. Furthermore, the control algorithms verified in this thesis could be deployed on a real-time embedded platform to conduct Hardware-in-the-Loop (HIL) tests, validating the computational efficiency of the MPC algorithm on target hardware.

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